Geometry-based Methods in Vision

CMU 16-822, Fall 2024

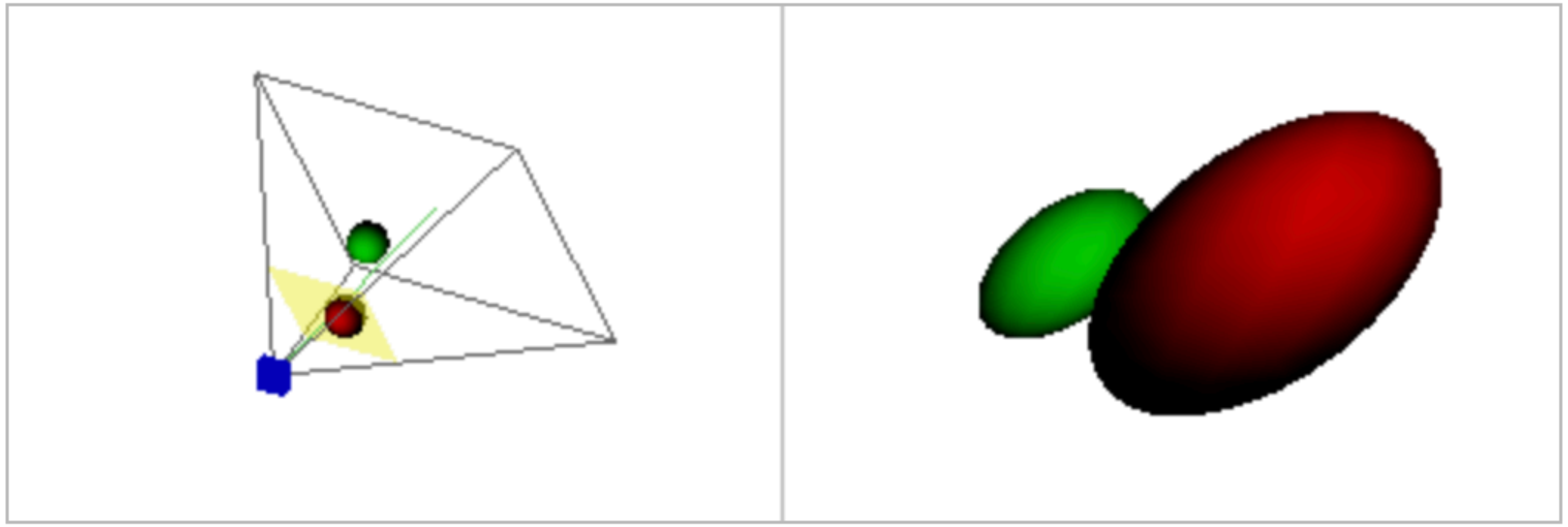
Instructor: Shubham Tulsiani

Administrivia

- PS1 out
  - *Due on Tue (5 days)*

- Assignment 1
  - *Due on Tue (week after)*

Lecture 6: Camera Models



A reminder to ask questions!

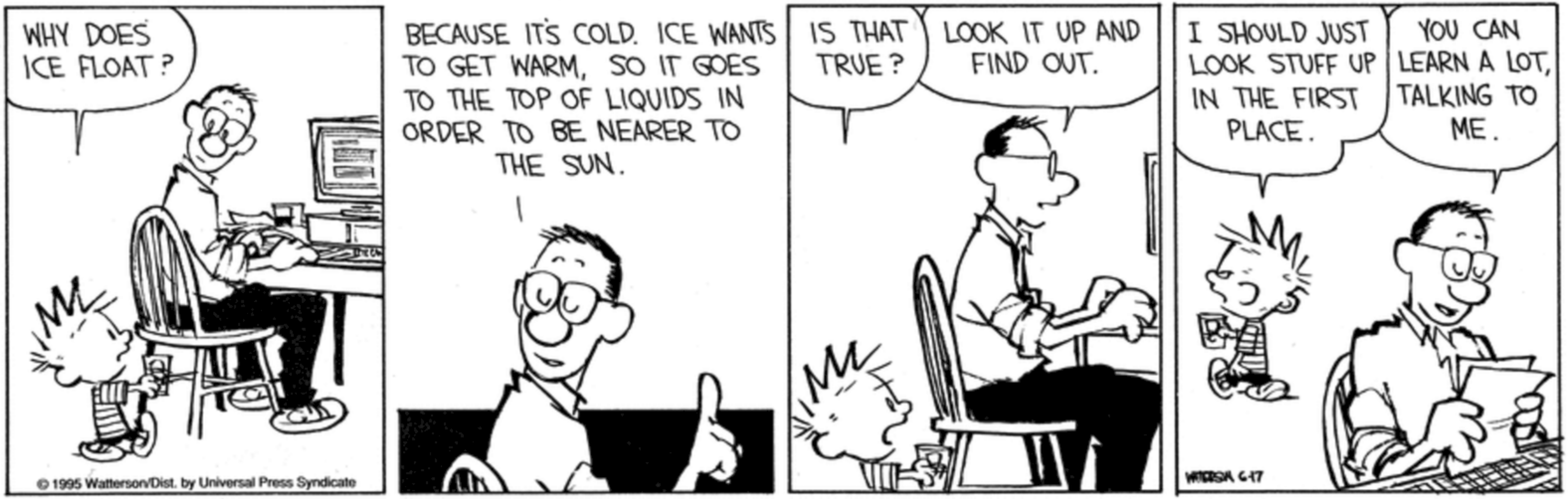


Image credits: Bill Watterson, “Calvin & Hobbes”

Beyond DLT

$$L = \sum_i d(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \qquad \min_H L$$

$$d_{\text{alg}}(\mathbf{x}, \mathbf{x}') = \|\mathbf{x}' \times \mathbf{x}\|^2$$

Need additional constraint to avoid degenerate solution

Closed form solution

Non-equivariant under similarity transform

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$

(Pixel projection loss)

No additional constraint needed

Not linear in **h**

Equivariant to similarity transform

Approach to using non-linear objectives:

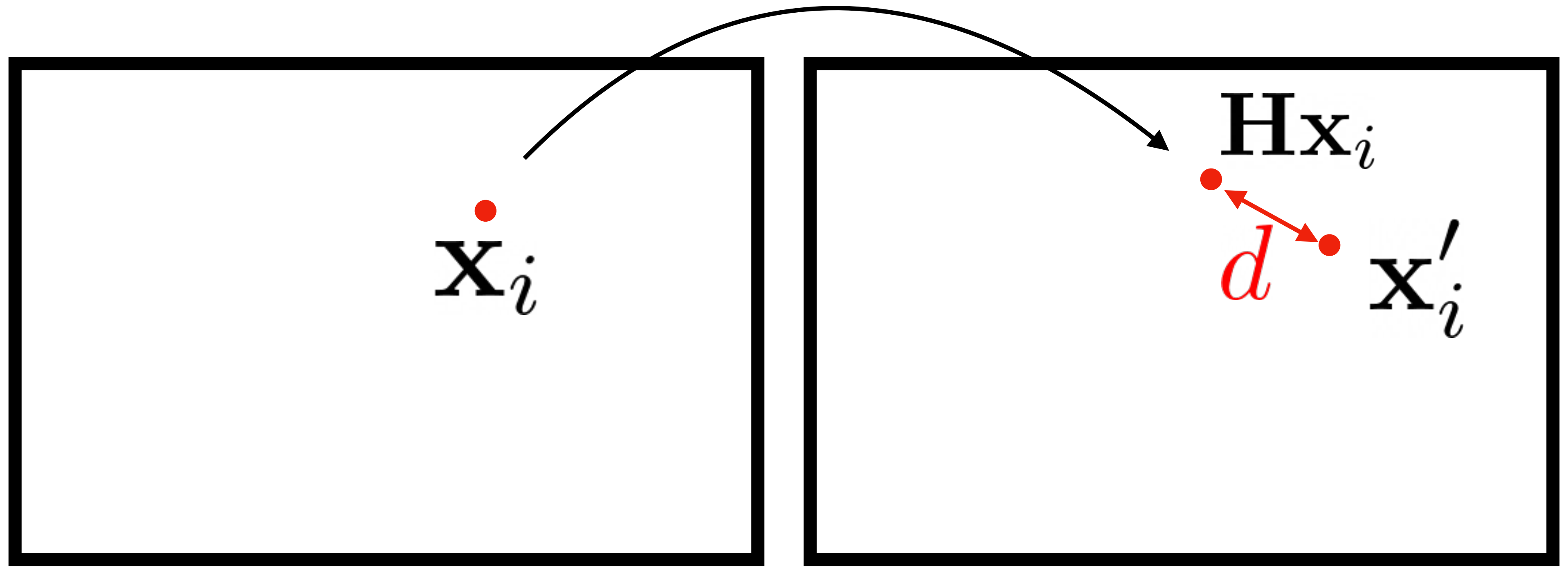
a) Initialize **H** using DLT

b) Iteratively update **H** using the non-linear objective

Beyond DLT

$$L = \sum_i d(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \qquad \min_H L$$

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$



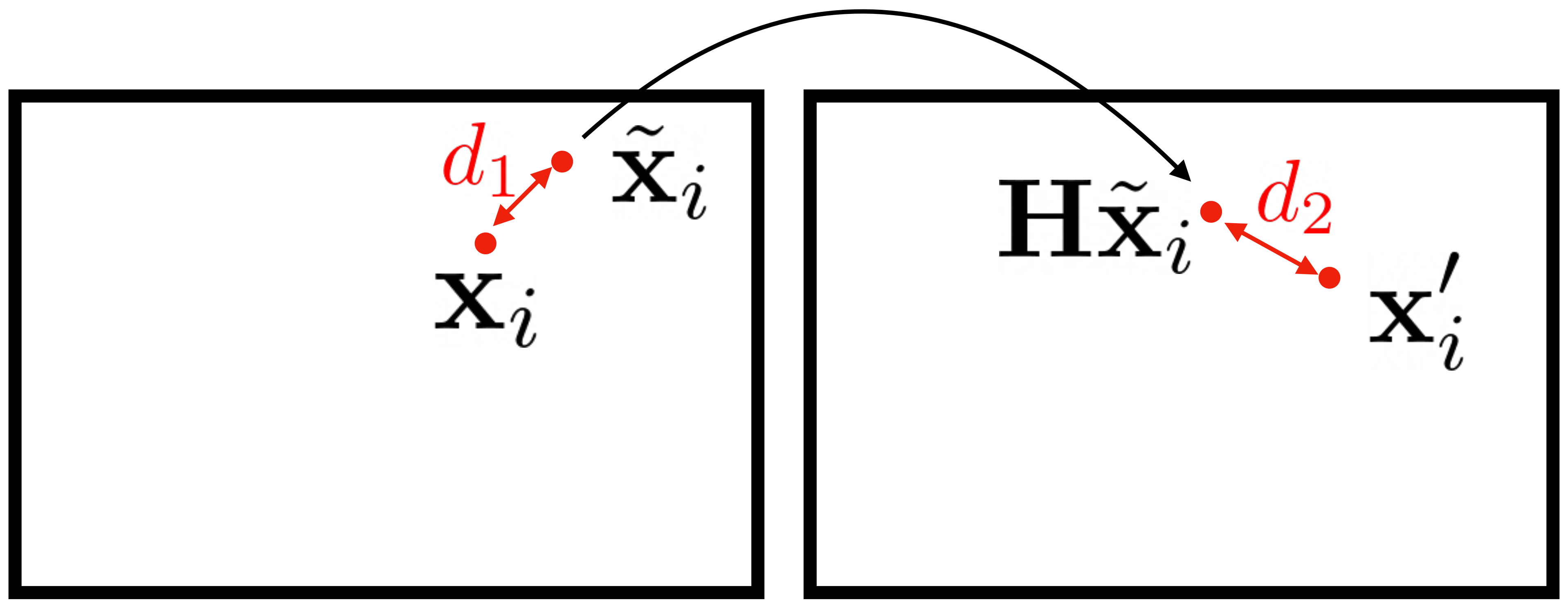
Only allows for error in one image!

Small annotation error in one image can cause much larger errors in other

Beyond DLT

$$L = \sum_i d(\tilde{\mathbf{x}}_i, \mathbf{x}_i) + \sum_i d(\mathbf{H}\tilde{\mathbf{x}}_i, \mathbf{x}'_i) \qquad \min_{H, \{\tilde{\mathbf{x}}_i\}} L$$

$$d_{\text{geom}}(\mathbf{x}, \mathbf{x}') = \|x'/w' - x/w\|^2 + \|y'/w' - y/w\|^2$$



Need to optimize 2n + 8 variables

Q: Do 4 correspondences still suffice?

Allow for error in both images

Summary

$$L = \sum_i d_{alg}(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \qquad \min_H L \quad \longrightarrow \quad \min_{\mathbf{h}, \|\mathbf{h}\|=1} \|A\mathbf{h}\|$$

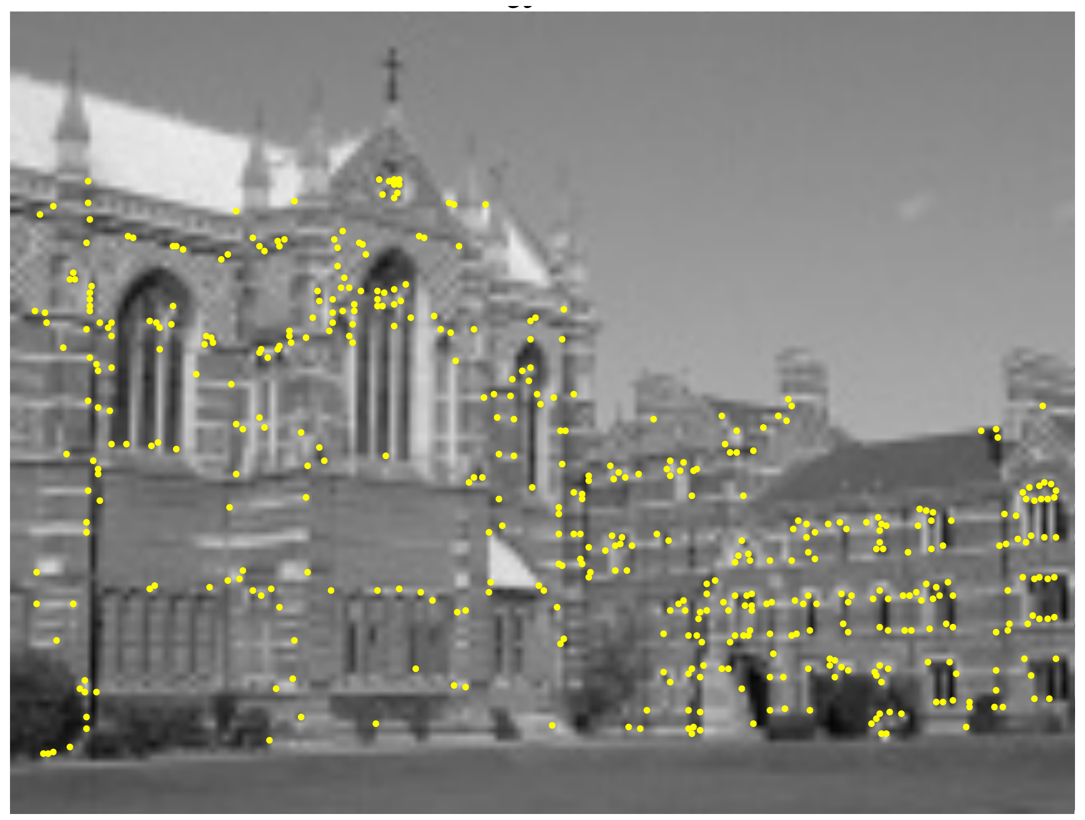
$$L = \sum_i d_{geom}(\mathbf{H}\mathbf{x}_i, \mathbf{x}'_i) \qquad \min_H L \quad \longrightarrow$$ Iterative optimization after initialization

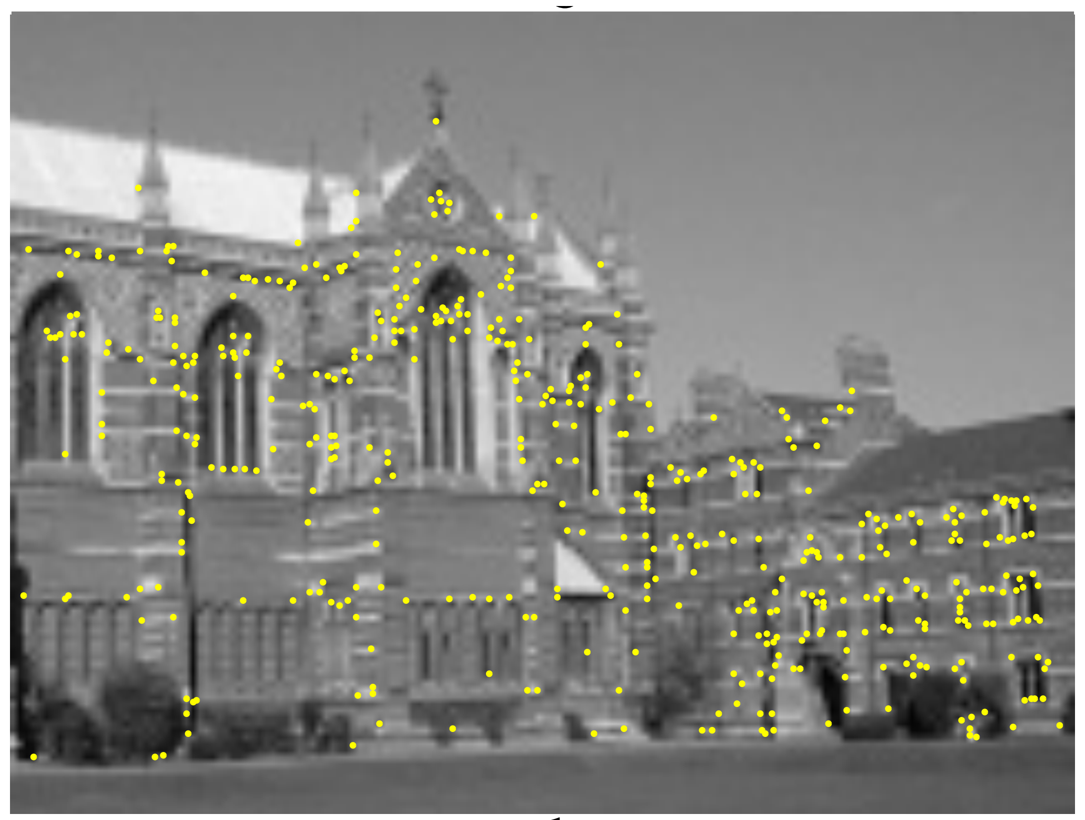
$$L = \sum_i d_g(\tilde{\mathbf{x}}_i, \mathbf{x}_i) + \sum_i d_g(\mathbf{H}\tilde{\mathbf{x}}_i, \mathbf{x}'_i) \qquad \min_{H, \{\tilde{\mathbf{x}}_i\}} L \quad \longrightarrow$$ Iterative optimization after initialization

Q: Which objective should I actually use?

Often, the first suffices (but normalization is important!)

Dealing with Noise



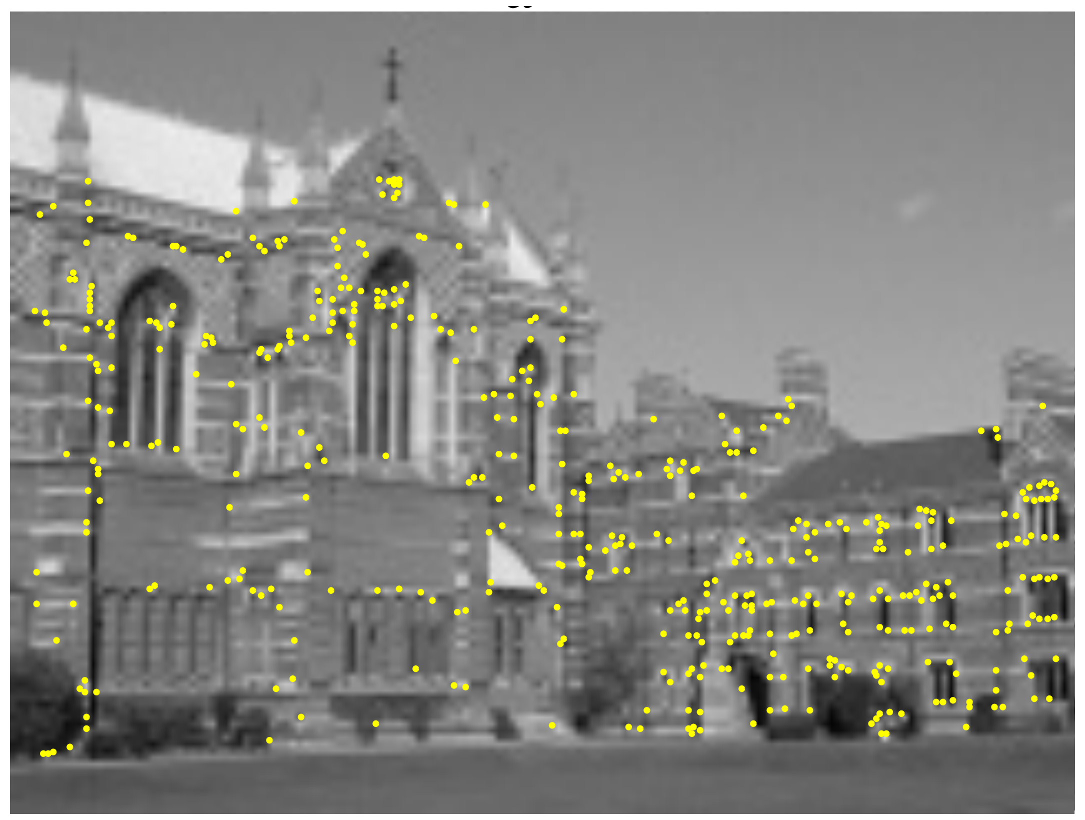
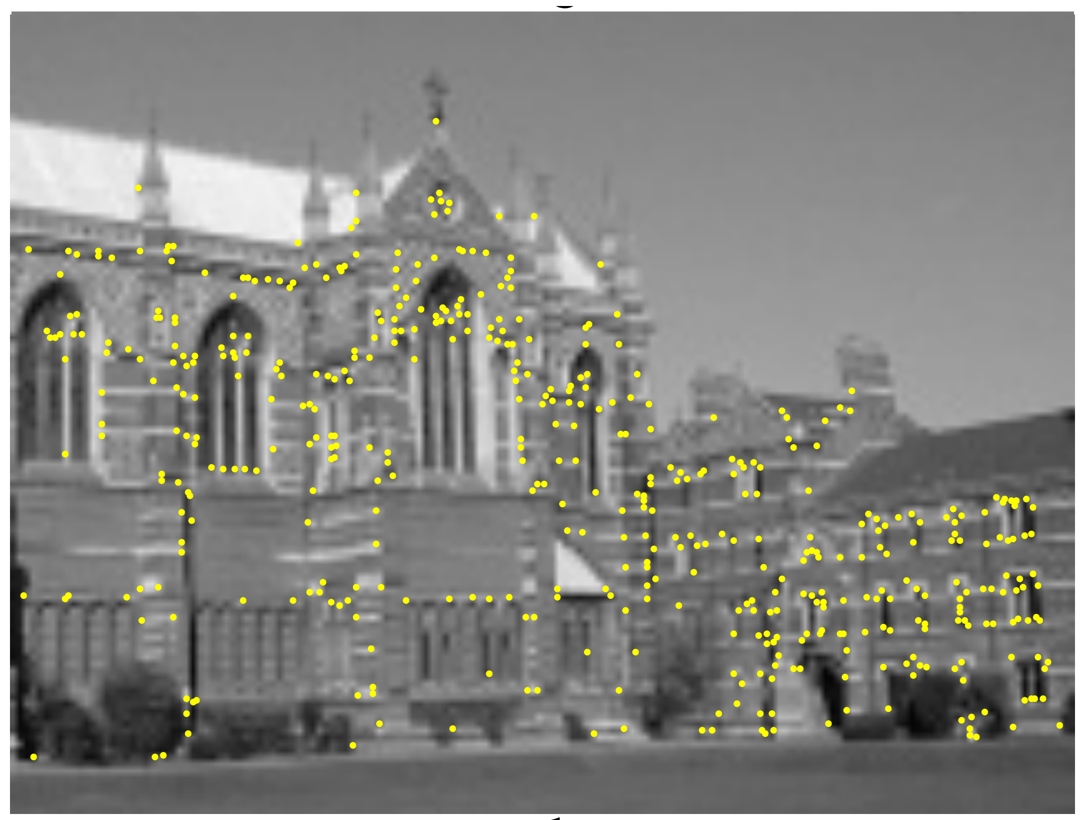


Step 1: Compute interest points and descriptors

Step 2: Find potential matches ($\mathbf{x_i}$, $\mathbf{x_i}$')

~~Step 3: Find optimal $\mathbf{H}$~~

Dealing with Noise: RANSAC

Step 1: Compute interest points and descriptors

Step 2: Find potential matches ($\mathbf{x_i}$, $\mathbf{x_i}$')

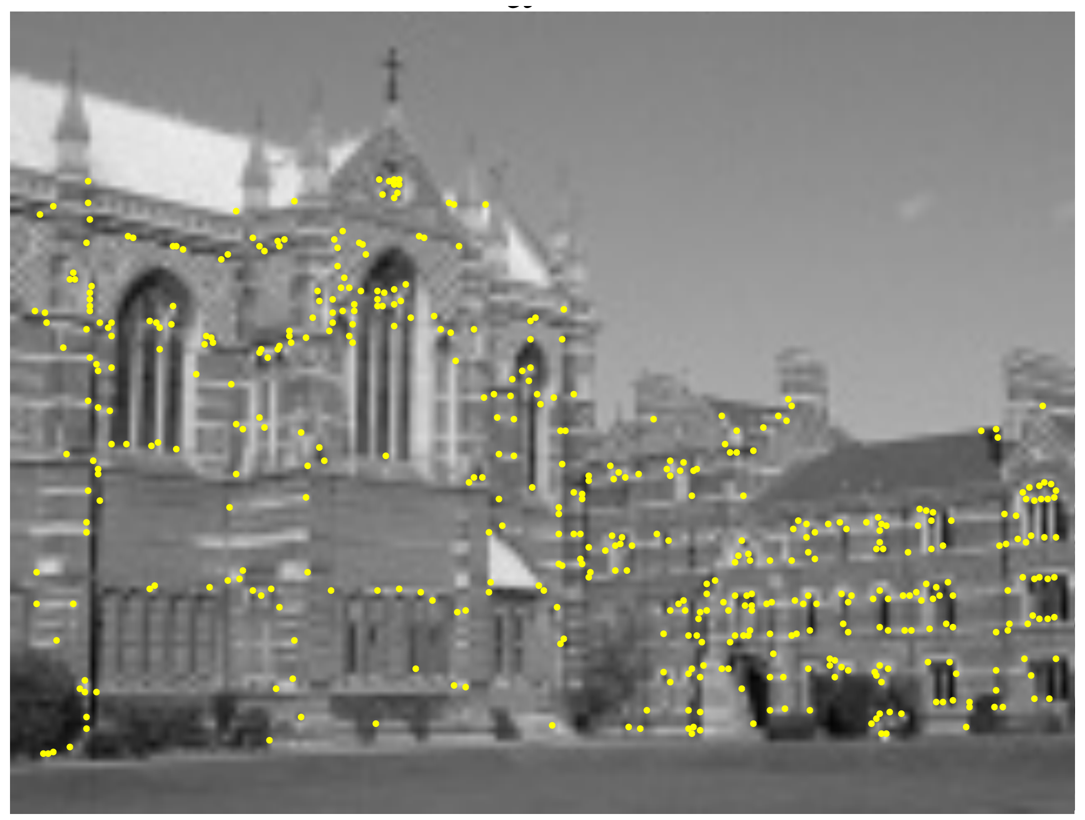
Step 3: For N iterations:

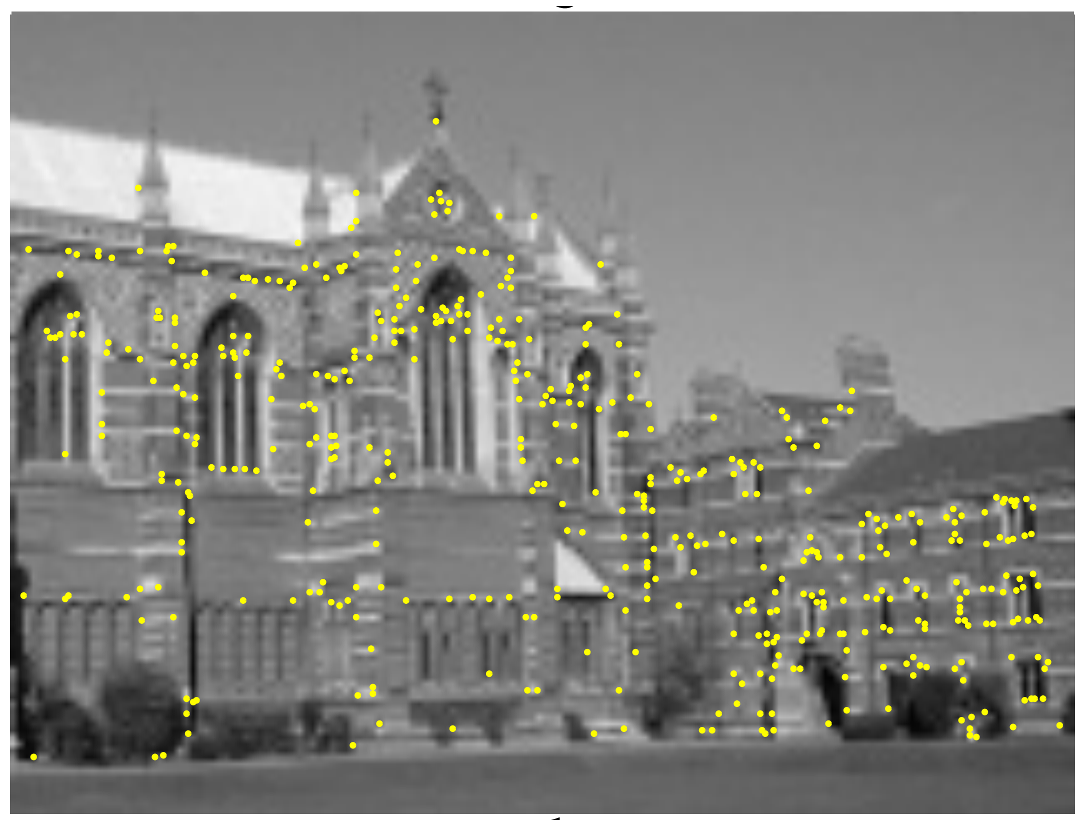
- Sample K potential matches to compute **H_n**
- Compute number of inliers (matches with error < *e* w.r.t. **H_n**)

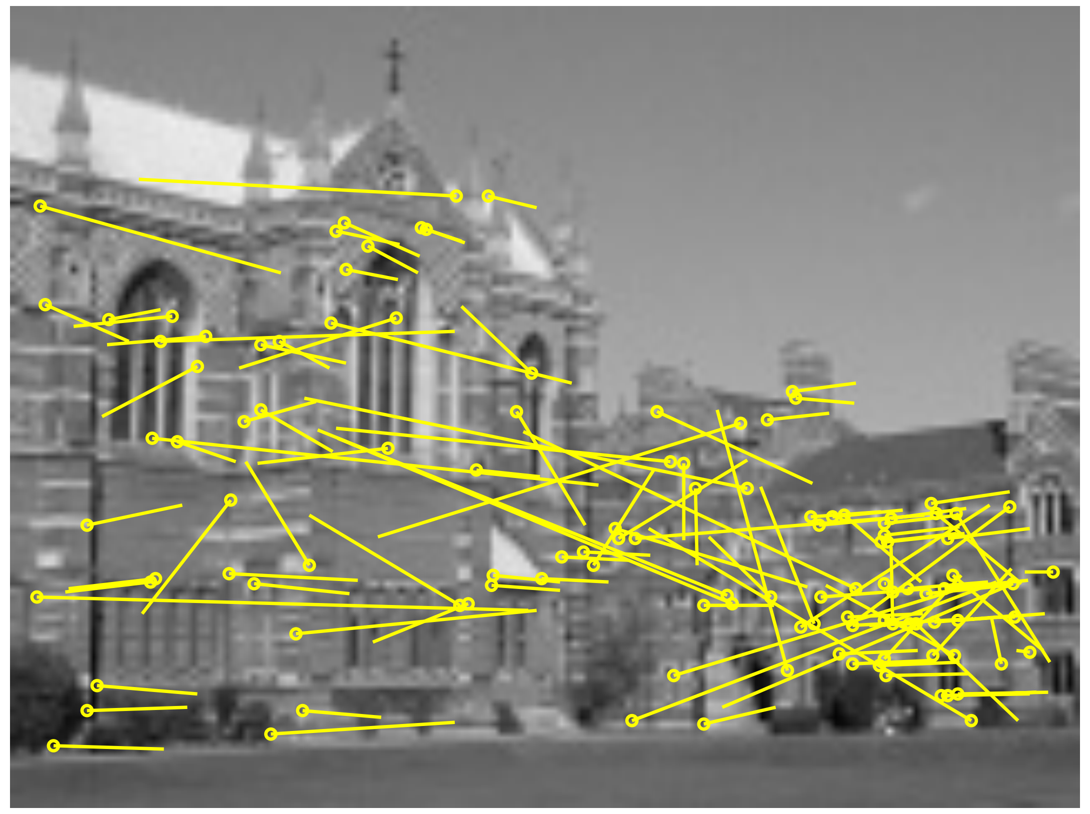
Several tunable hyper-parameters

Use the largest set of inliers to compute **H**

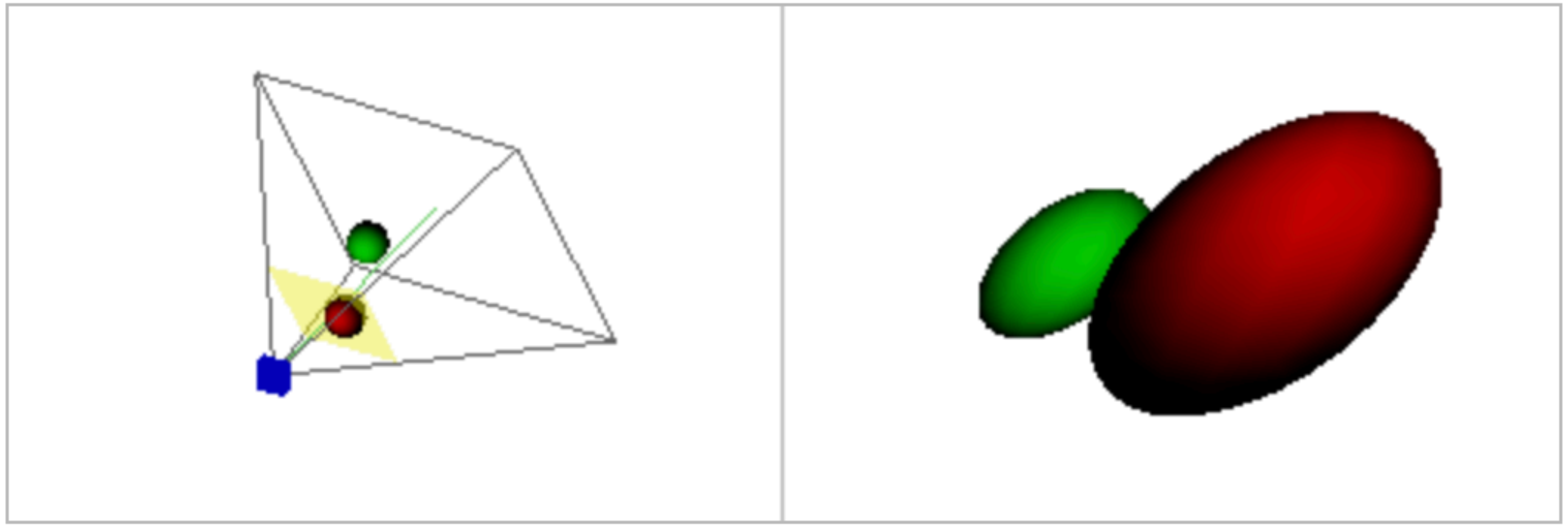
Dealing with Noise: RANSAC







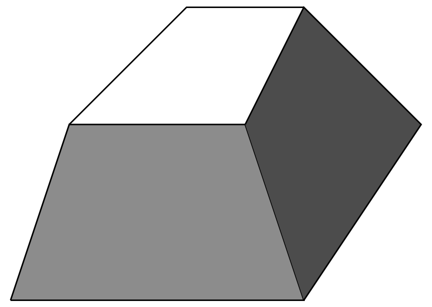
Lecture 6: Camera Models



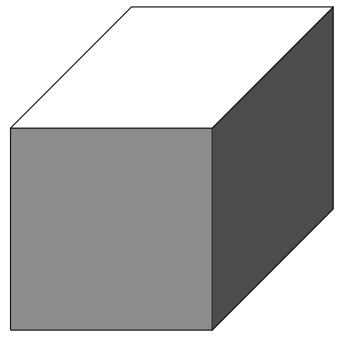
Projective Transforms from 3D to 2D: Cameras!

$$\mathbf{x}' = \mathrm{H}\mathbf{x}$$

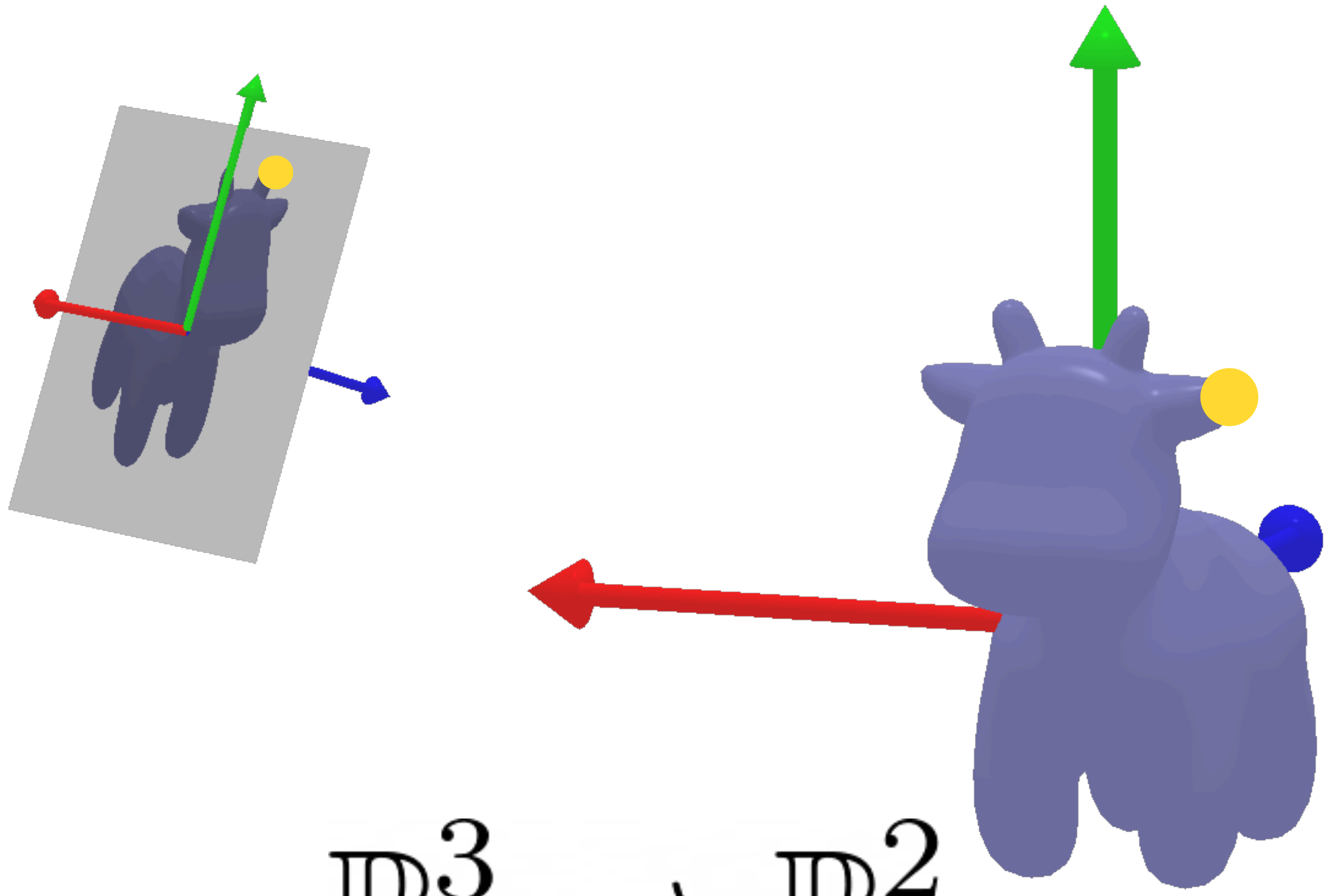
$$\mathbb{P}^2 \to \mathbb{P}^2$$



$$\mathbf{x}' = \mathrm{H}\mathbf{x}$$

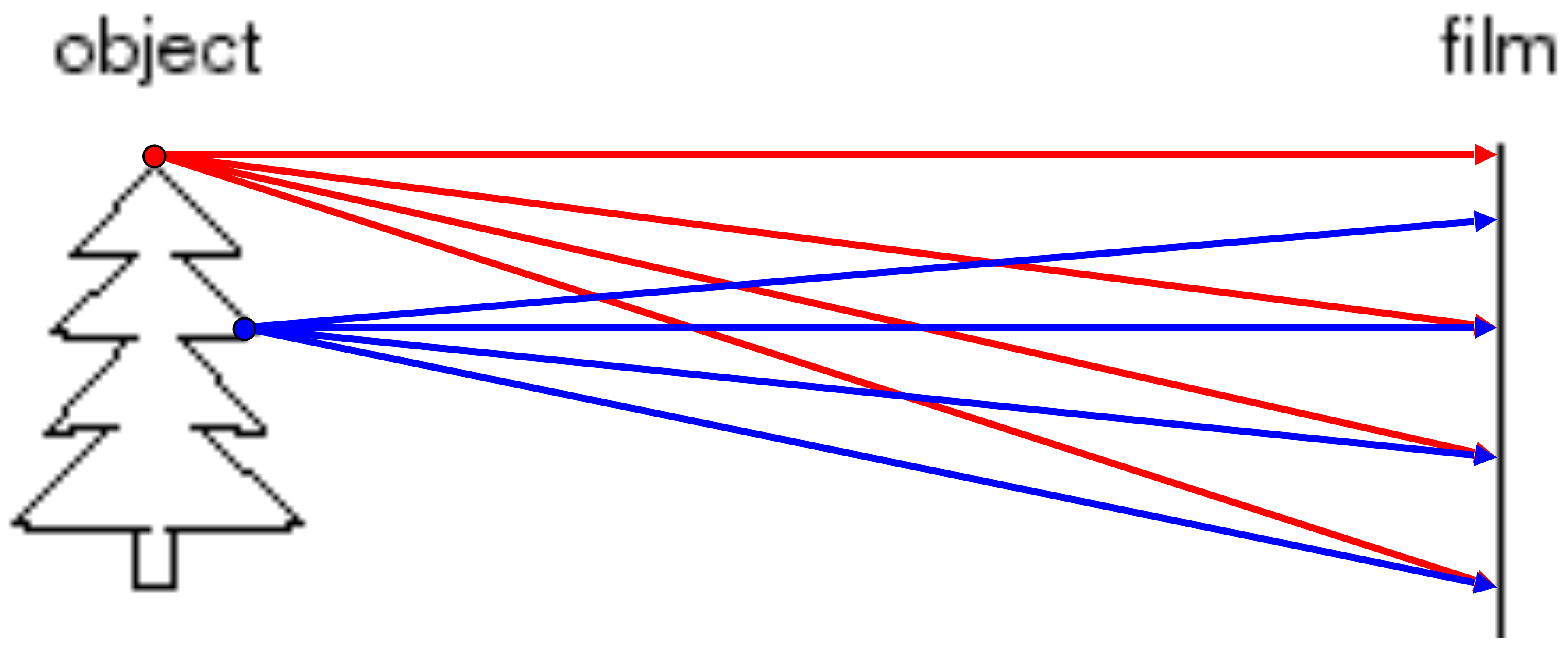


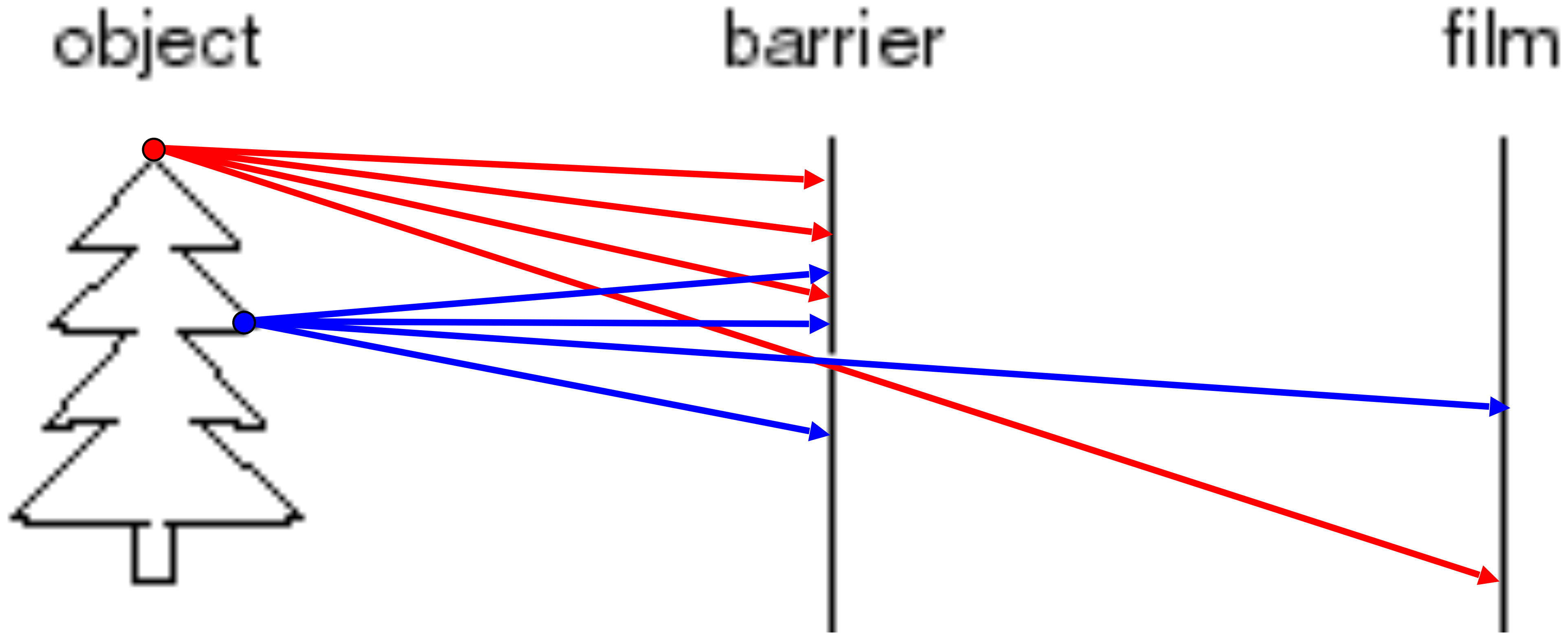
$$\mathbb{P}^3 \to \mathbb{P}^3$$



$$\mathbb{P}^3 \to \mathbb{P}^2$$

Let's make a Camera!





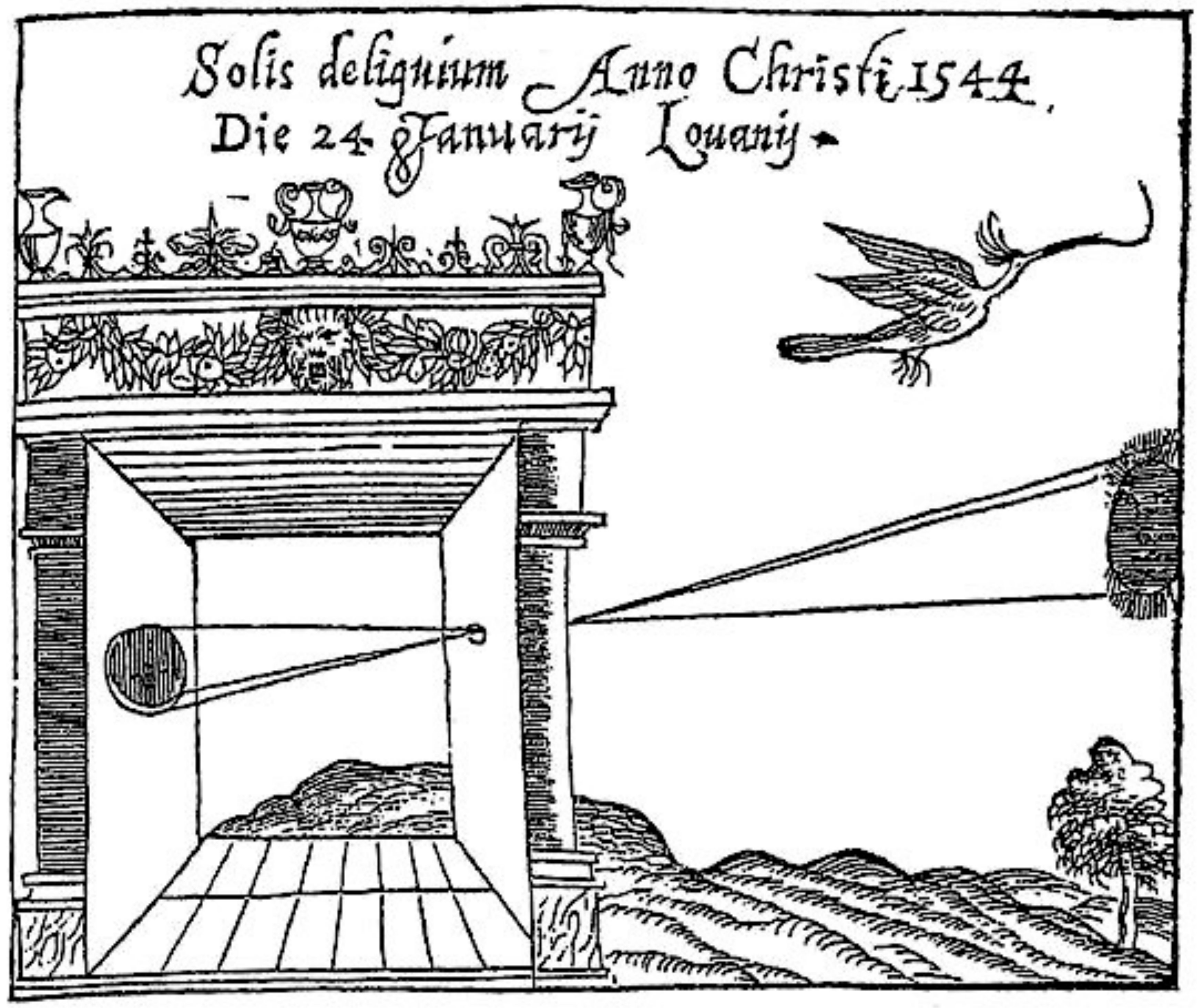
Idea 1: Put piece of film in front of an object

Add a barrier to block most rays.

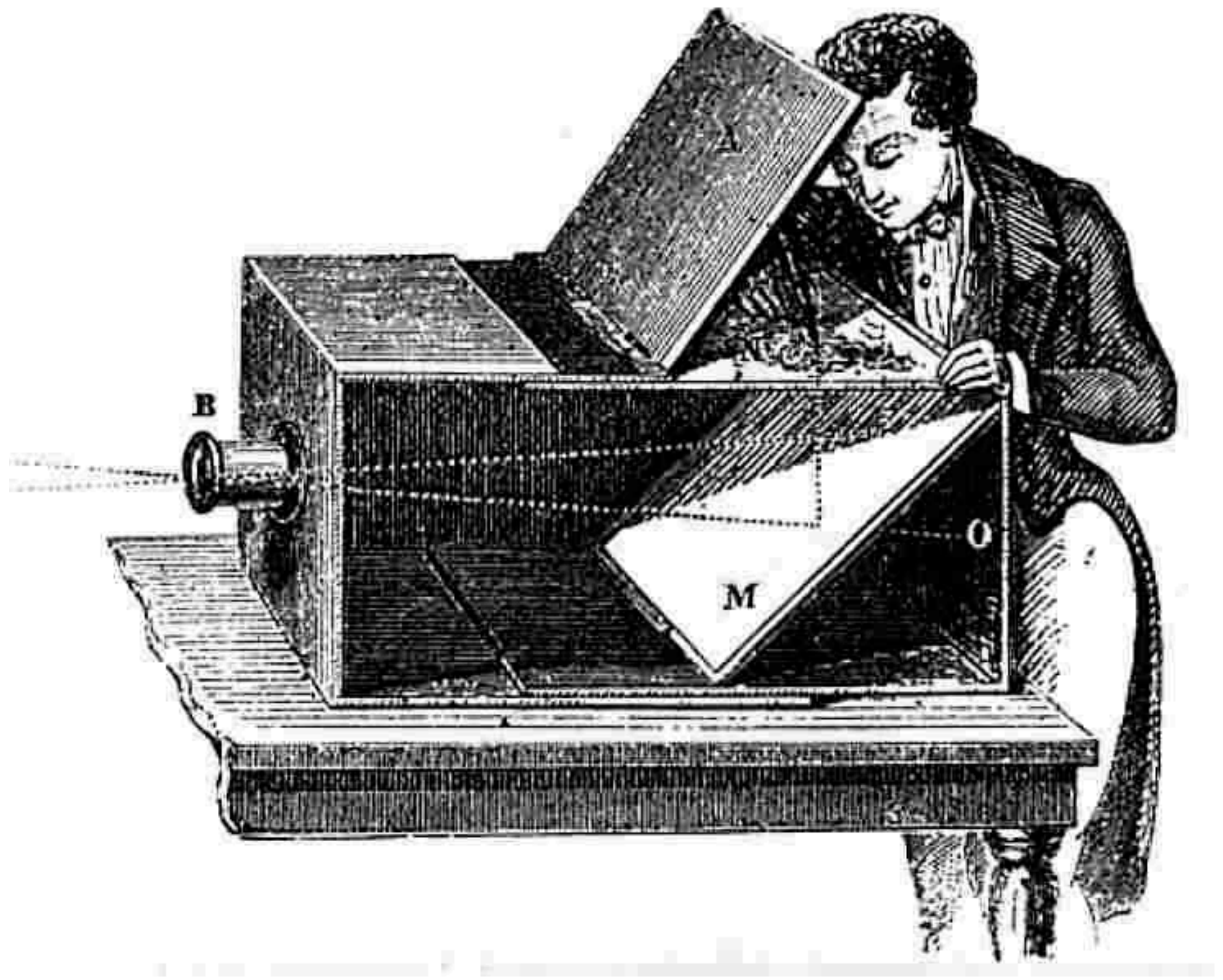
Slide credit: Steve Seitz

Camera Obscura

- Basic principle known to Mozi (470-390 BCE), Aristotle (384-322 BCE)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)



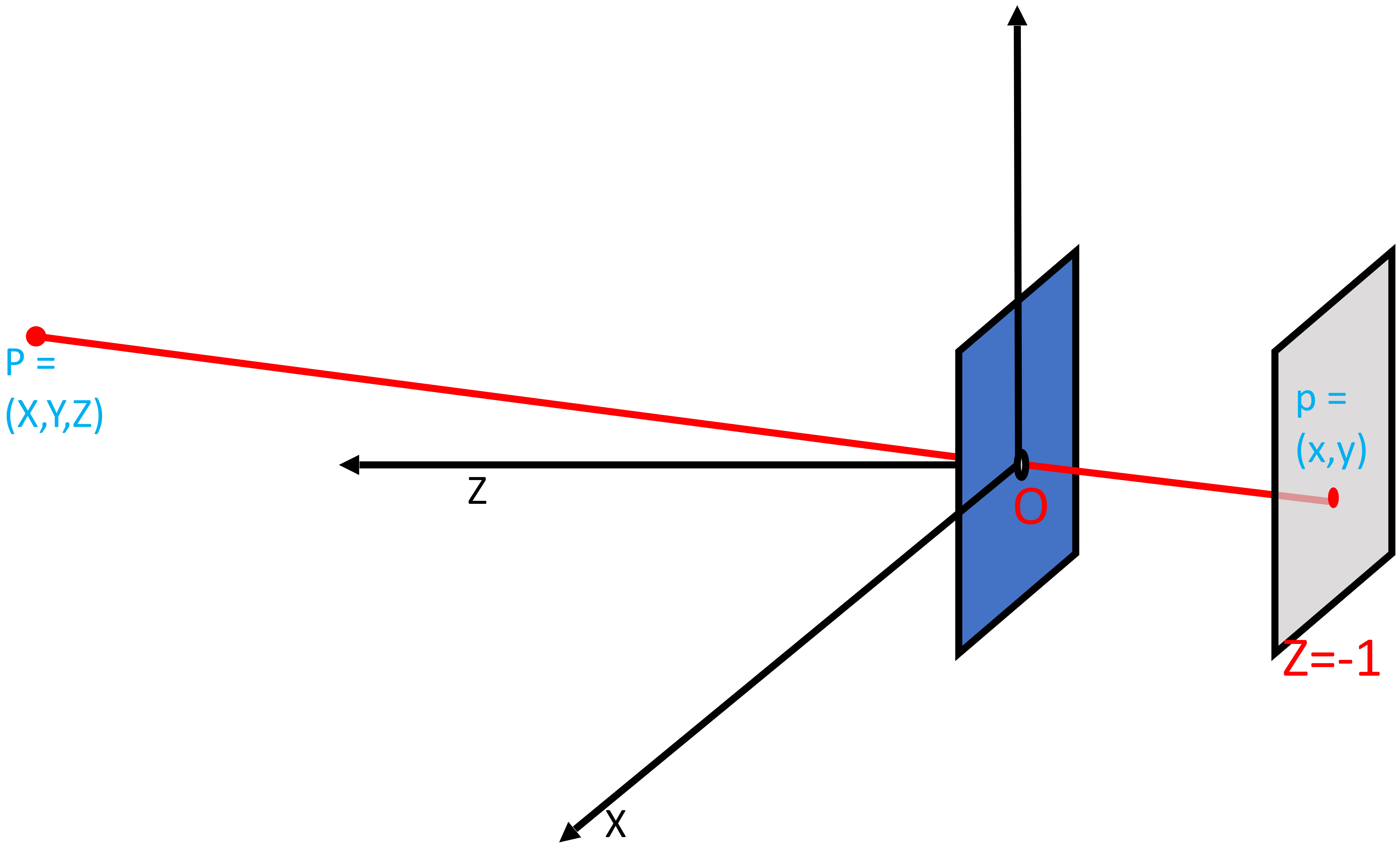
Gemma Frisius, 1558



Slide credit: Lana Lazebnik

Image source

Pinhole Projection Model

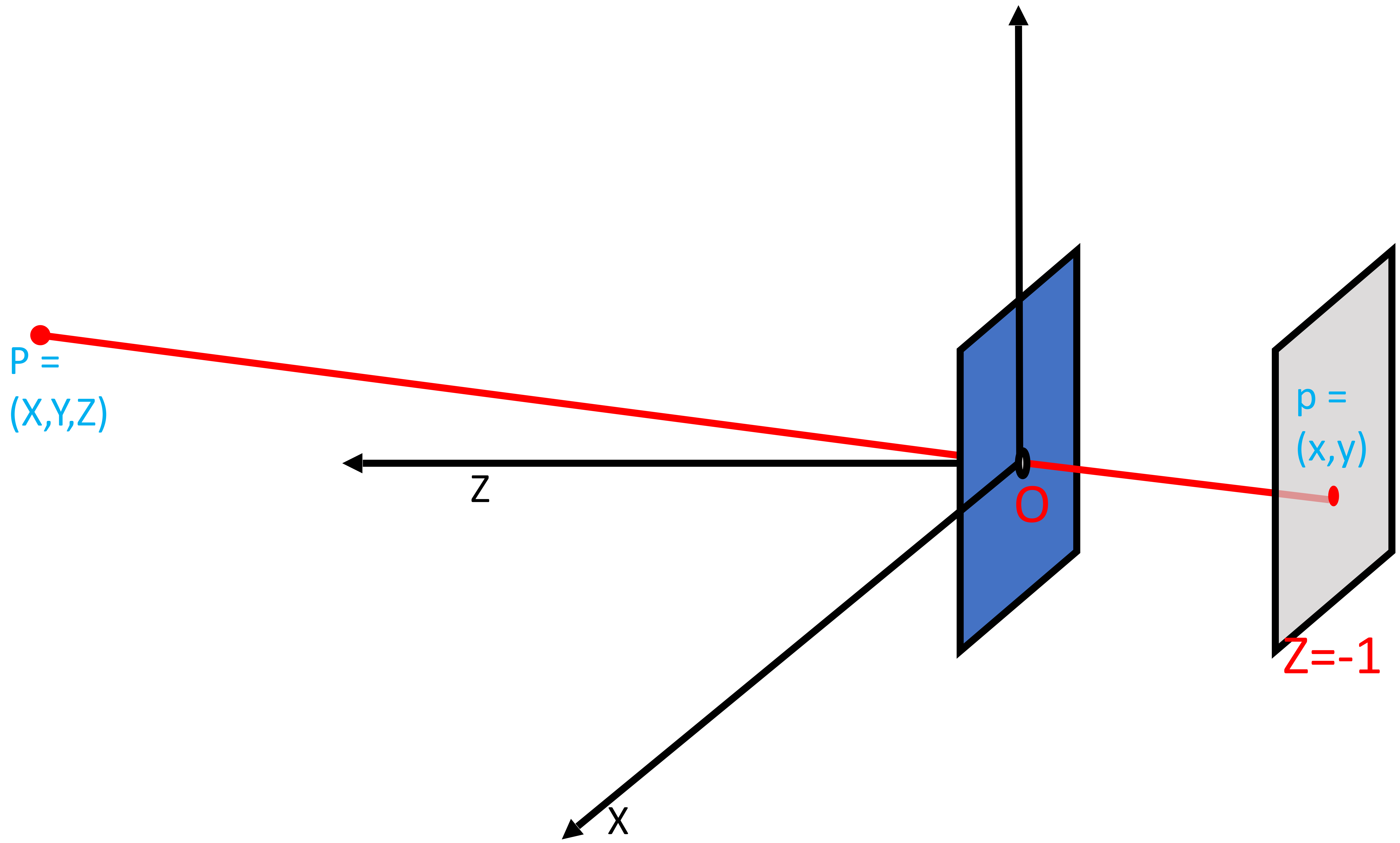


$$x = \frac{-X}{Z}$$

$$y = \frac{-Y}{Z}$$

Slide credit: Bharath Hariharan

Pinhole Projection Model



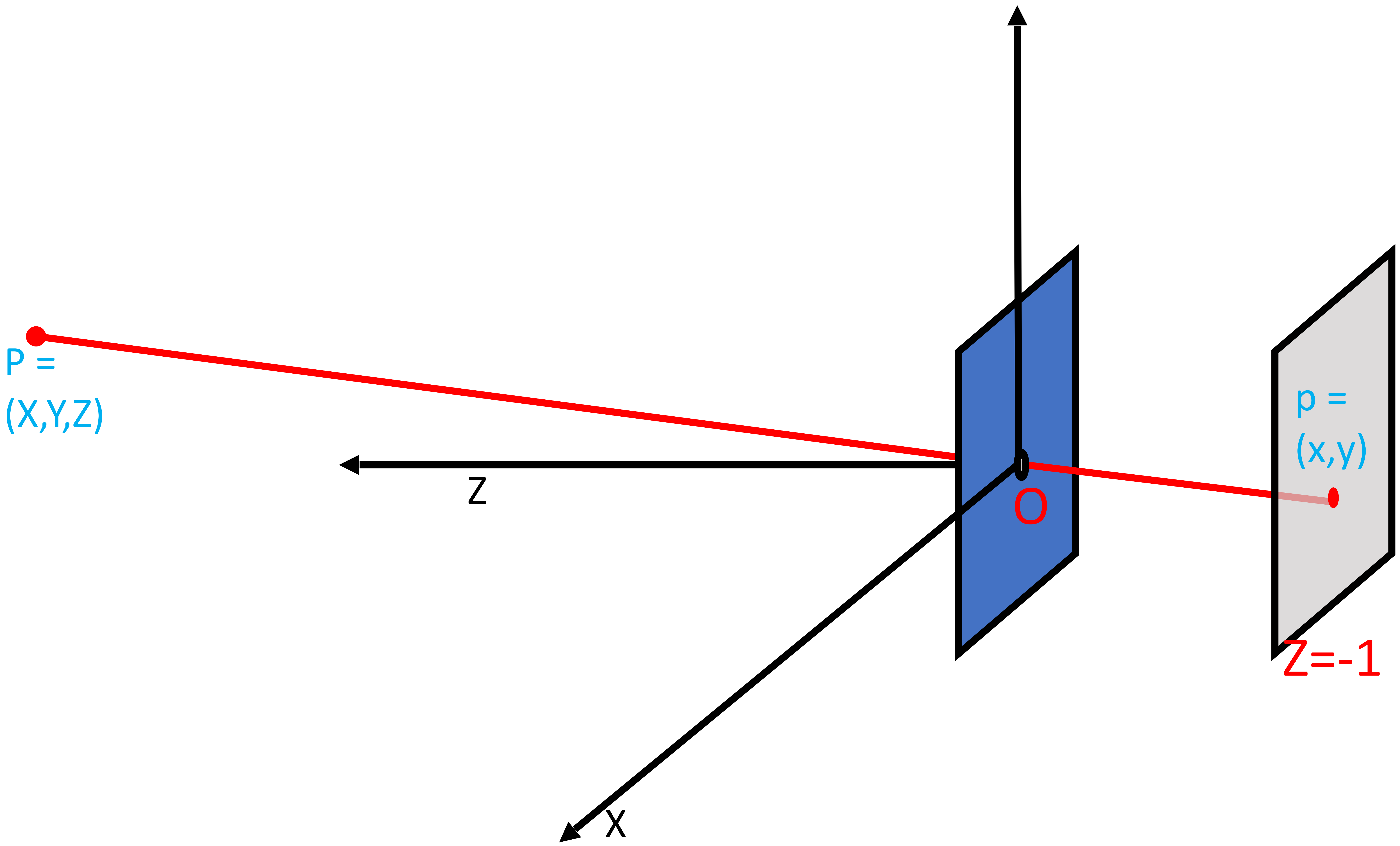
$$x = \frac{X}{Z}$$

$$y = \frac{Y}{Z}$$

A more convenient convention

Slide credit: Bharath Hariharan

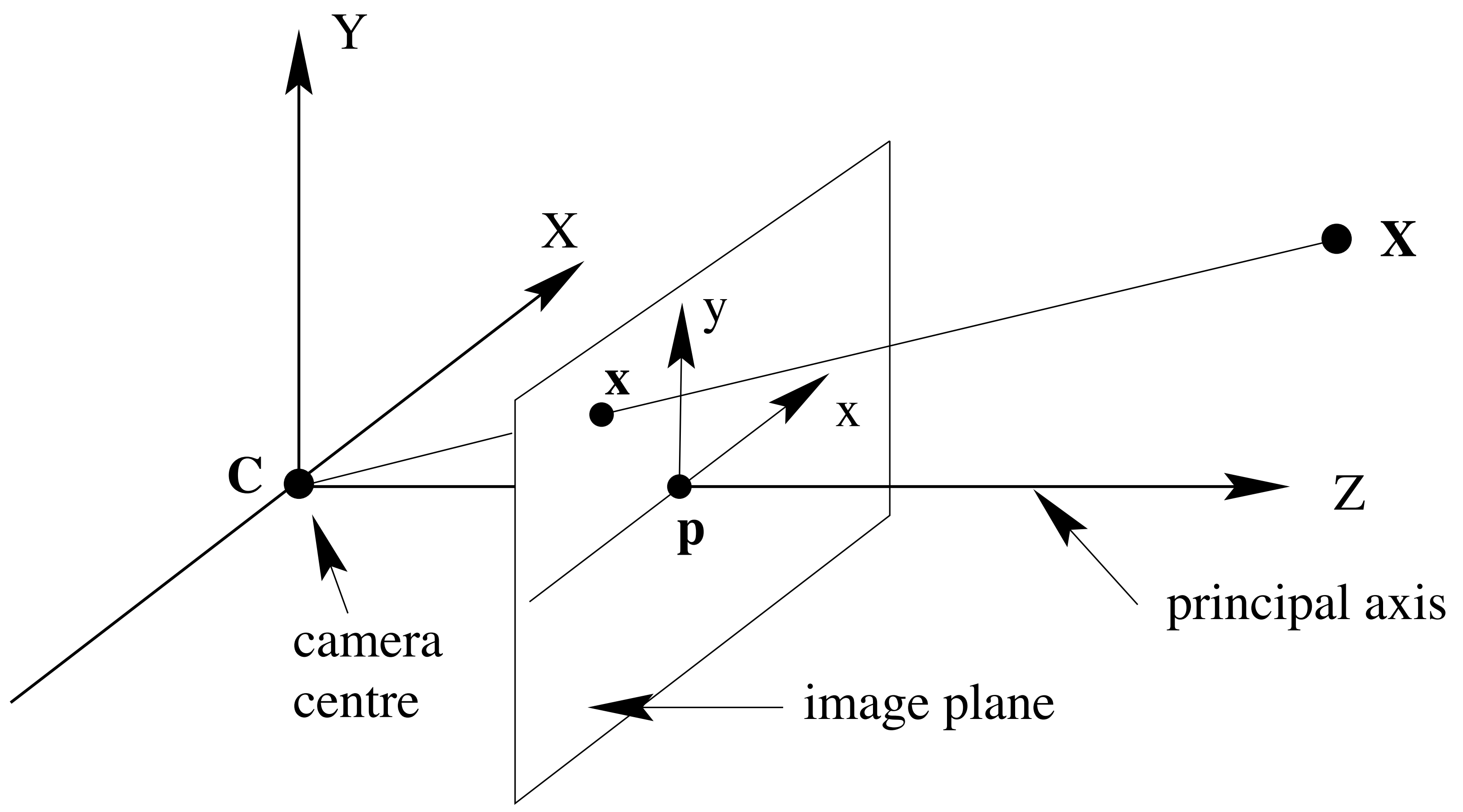
Pinhole Projection Model



$$\mathbf{x} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \mathbf{X}$$

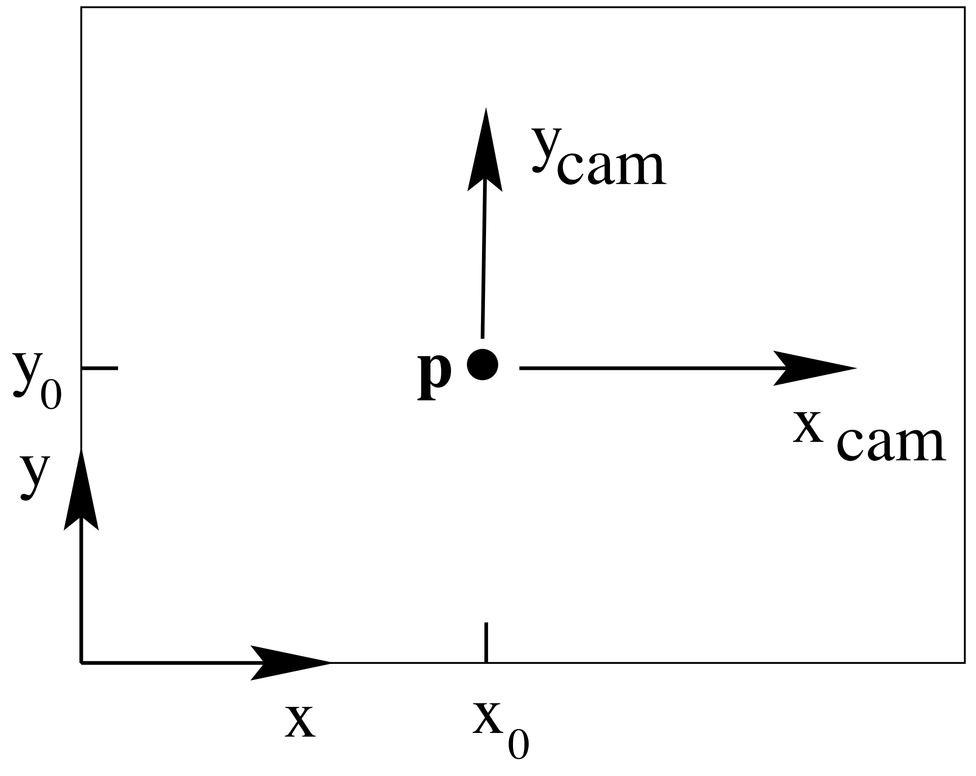
$\mathbb{P}^2$ $\mathbb{P}^3$

Focal length and Principal Point



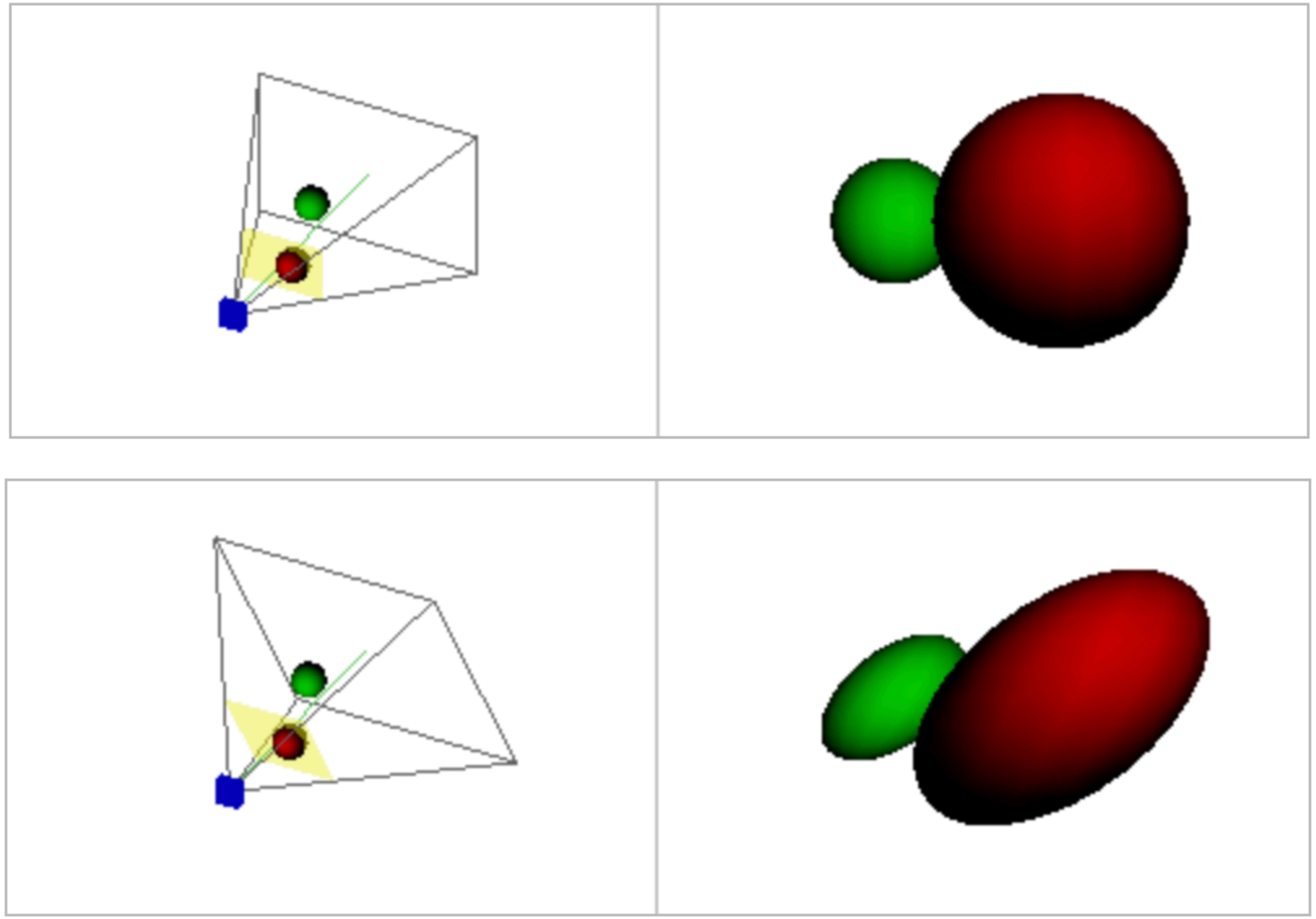
$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

$$P = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$



$$P = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

Camera Shear



Pixels correspond to parallelograms in the image plane

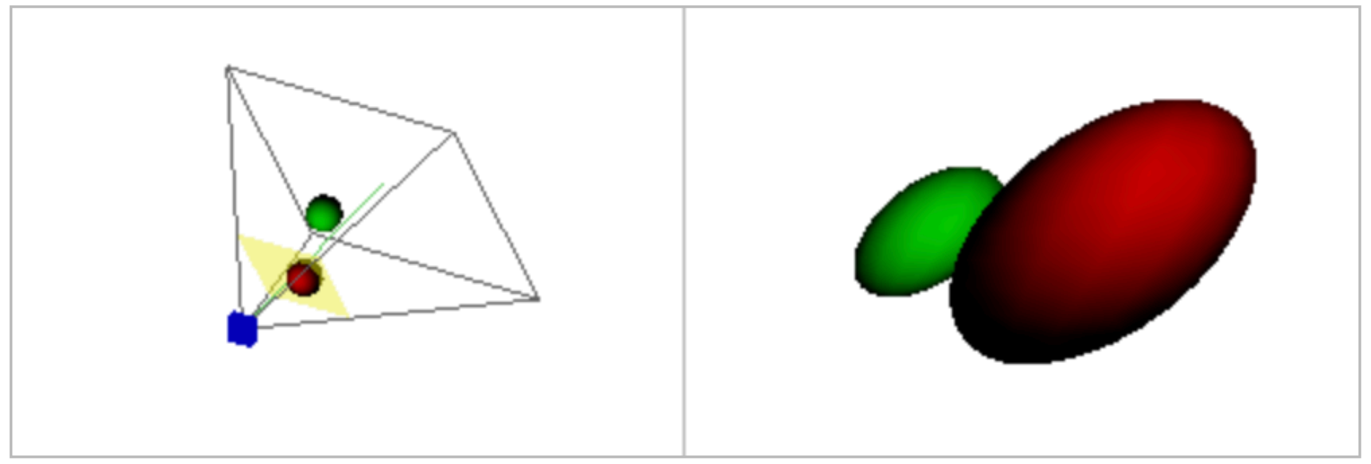
$$\mathbf{x} = \mathbf{P}\mathbf{X}$$

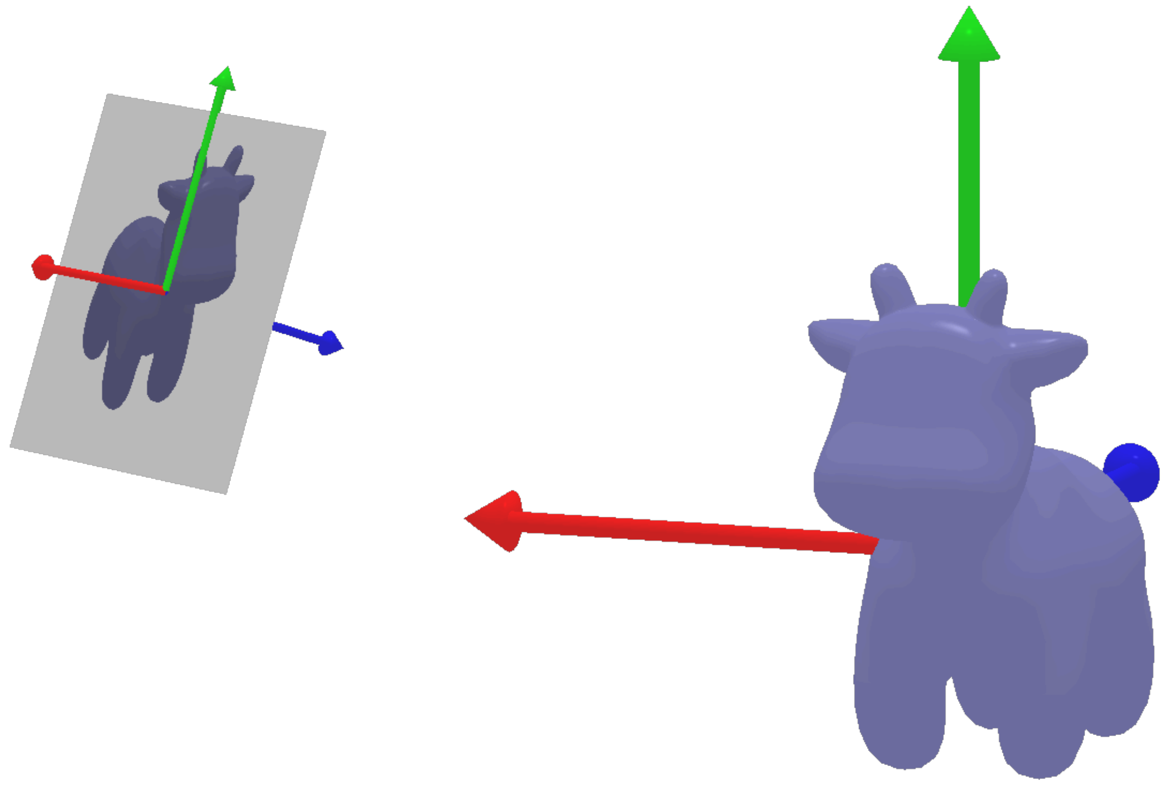
$$P = \begin{bmatrix} f & s & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

$$P = \mathbf{K} \begin{bmatrix} I & \mathbf{0} \end{bmatrix}$$

Image credit: https://ksimek.github.io/2013/08/13/intrinsic/

Rotation and Translation





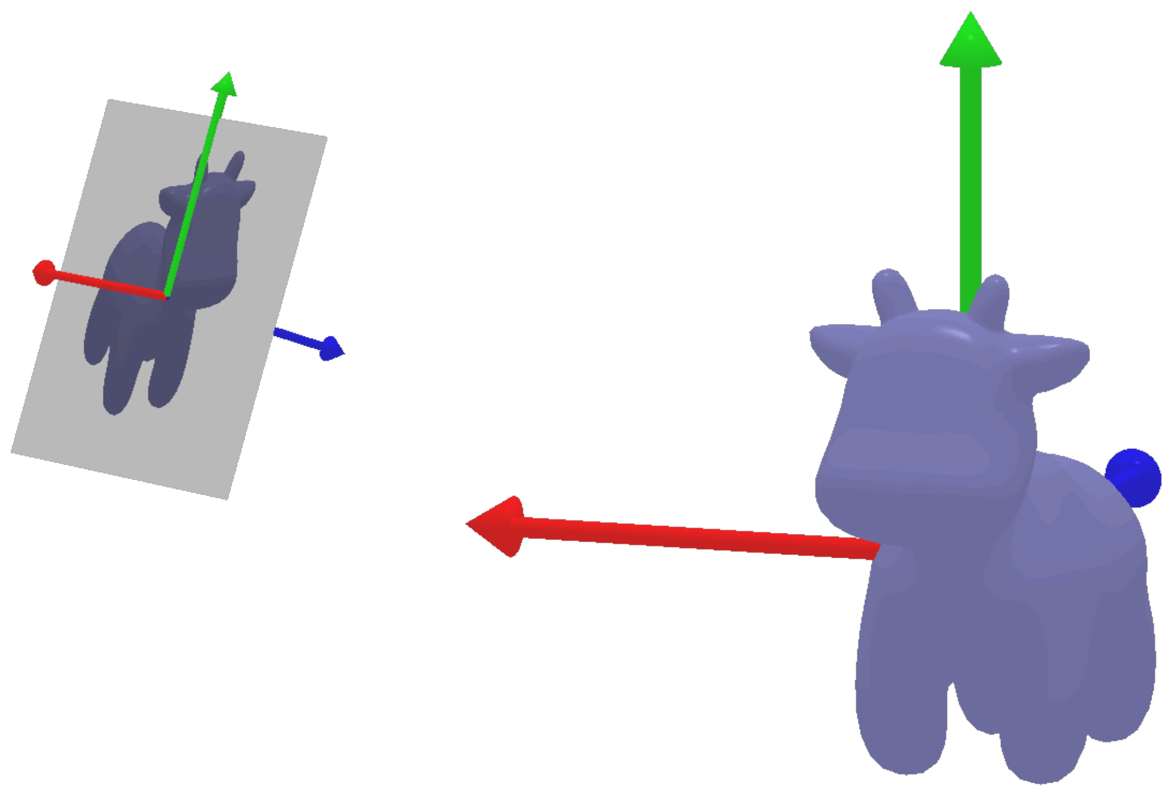
$$\mathbf{x} = \mathbf{K}\begin{bmatrix} I & \mathbf{0} \end{bmatrix} \mathbf{X}_{cam}$$

(inhomogeneous) Camera centre in world frame

$$\mathbf{X}_{cam} = \begin{bmatrix} R & -R\tilde{\mathbf{C}} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{X}_w$$

Image credit: https://ksimek.github.io/2013/08/13/intrinsic/

Camera Matrix



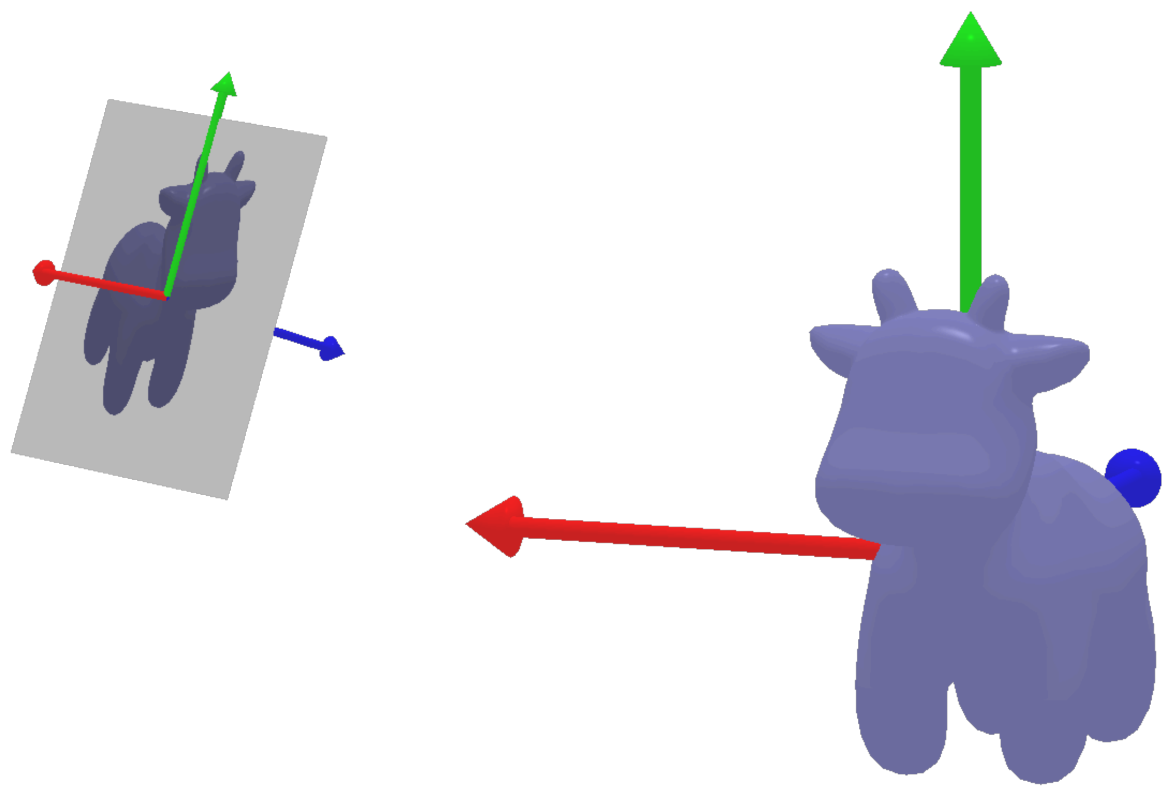
$$\mathbf{x} = \mathbf{P}\mathbf{X}_w$$

$$\mathbf{P} = \underline{\mathbf{K}}\begin{bmatrix} I & \mathbf{0} \end{bmatrix} \underline{\begin{bmatrix} R & -R\tilde{\mathbf{C}} \\ \mathbf{0} & 1 \end{bmatrix}}$$

Intrinsic Matrix: Relates 3D in camera frame to 2D image

Extrinsic Matrix: Relates 3D in world frame to 3D in camera frame

Camera Matrix



$$\mathbf{x} = \mathbf{P}\mathbf{X}_w$$

$$\mathbf{P} = \mathbf{K}\begin{bmatrix} I & \mathbf{0} \end{bmatrix}\begin{bmatrix} R & -R\tilde{\mathbf{C}} \\ \mathbf{0} & 1 \end{bmatrix}$$

$$\mathbf{P} = \mathbf{K}R\begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

Q: How many degrees of freedom?

Q: Can any projection matrix **P** be decomposed this way?

$$\mathbf{P} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix} \qquad \mathbf{M} = \mathbf{K}R \qquad \tilde{\mathbf{C}} = -\mathbf{M}^{-1}\mathbf{p}_4$$

Yes, assuming M is rank 3 equivalently, camera centre is not at infinity!

Two views of a Camera Matrix

$$\mathbf{P} = \mathbf{K}R\left[I \quad -\tilde{\mathbf{C}}\right]$$

Geometrically motivated view,
disentangling extrinsics and intrinsics

$$\mathbf{P} = \left[\mathbf{M} \quad \mathbf{p}_4\right]$$

A general projection from 3D to 2D

Q: What should be the rank of P to form a valid image?

Q: What happens if: rank = 2? rank = 1?

mapping is constrained to a line (or a single point)

Two views of a Camera Matrix

$$\mathbf{P} = \mathbf{K}R\left[I \quad -\tilde{\mathbf{C}}\right]$$

$$\mathbf{P} = \left[\mathbf{M} \quad \mathbf{p}_4\right]$$

Camera Centre

$$\mathbf{C} \equiv \begin{bmatrix} \tilde{\mathbf{C}} \\ 1 \end{bmatrix}$$

$$\mathbf{C} \text{ s.t. } \mathbf{PC} = \mathbf{0}$$

Q: How would you compute this given **P**?

Two views of a Camera Matrix

$$\mathbf{P} = \mathbf{K}R\begin{bmatrix} I & -\tilde{\mathbf{C}} \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix}$$

Camera Centre

$$\mathbf{C} \equiv \begin{bmatrix} \tilde{\mathbf{C}} \\ 1 \end{bmatrix}$$

$$\mathbf{C} \text{ s.t. } \mathbf{PC} = \mathbf{0}$$

Show that all (3D) points on a line joining camera centre and a point X project to the same (2D) point

can intuitively see this in
the geometric view ..

$$\mathbf{P}(\alpha\mathbf{X} + \beta\mathbf{C})$$

$$= \alpha\mathbf{PX} = \mathbf{PX}$$

Properties of a Camera Matrix

$$\mathbf{P} = \begin{bmatrix} \mathbf{M} & \mathbf{p}_4 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{p}^{1\mathsf{T}} \\ \mathbf{p}^{2\mathsf{T}} \\ \mathbf{p}^{3\mathsf{T}} \end{bmatrix} = \begin{bmatrix} \mathbf{m}^{1\mathsf{T}} & p_{14} \\ \mathbf{m}^{2\mathsf{T}} & p_{24} \\ \mathbf{m}^{3\mathsf{T}} & p_{34} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 & \mathbf{p}_4 \end{bmatrix}$$

Camera Centre

$$\mathbf{C} \text{ s.t. } \mathbf{PC} = \mathbf{0}$$

Projection of (3D) points on a line joining camera centre and a point X

$$\mathbf{P}(\alpha\mathbf{X} + \beta\mathbf{C}) = \alpha\mathbf{PX} = \mathbf{PX}$$

Principal Plane

All points s.t. their projection maps to the line at infinity

$$\mathbf{PX} \sim \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

$$\pi \equiv \mathbf{p}^3$$

Properties of a Camera Matrix

$$\mathbf{P} = [\mathbf{M} \quad \mathbf{p}_4]$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{p}^{1\mathsf{T}} \\ \mathbf{p}^{2\mathsf{T}} \\ \mathbf{p}^{3\mathsf{T}} \end{bmatrix} = \begin{bmatrix} \mathbf{m}^{1\mathsf{T}} & p_{14} \\ \mathbf{m}^{2\mathsf{T}} & p_{24} \\ \mathbf{m}^{3\mathsf{T}} & p_{34} \end{bmatrix} = [\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4]$$

Camera Centre

$$\mathbf{C} \text{ s.t. } \mathbf{PC} = \mathbf{0}$$

Projection of (3D) points on a line joining camera centre and a point X

$$\mathbf{P}(\alpha\mathbf{X} + \beta\mathbf{C}) = \alpha\mathbf{PX} = \mathbf{PX}$$

Principal Plane

$$\pi \equiv \mathbf{p}^3$$

Unprojection What points **X** can project to a coordinate **u**?

$$\mathbf{P}^+ = \mathbf{P}^\mathsf{T}(\mathbf{PP}^\mathsf{T})^{-1}$$

Q: Why is this well-defined?

$\mathbf{P}^+\mathbf{u}$ projects to **u**

$\lambda\mathbf{P}^+\mathbf{u} + \mathbf{C}$ projects to **u**

Properties of a Camera Matrix

$$\mathbf{P} = \begin{bmatrix}\mathbf{M} & \mathbf{p}_4\end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix}\mathbf{p}^{1\mathsf{T}} \\ \mathbf{p}^{2\mathsf{T}} \\ \mathbf{p}^{3\mathsf{T}}\end{bmatrix} = \begin{bmatrix}\mathbf{m}^{1\mathsf{T}} & p_{14} \\ \mathbf{m}^{2\mathsf{T}} & p_{24} \\ \mathbf{m}^{3\mathsf{T}} & p_{34}\end{bmatrix} = \begin{bmatrix}\mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 & \mathbf{p}_4\end{bmatrix}$$

Camera Centre

$$\mathbf{C} \text{ s.t. } \mathbf{PC} = \mathbf{0}$$

Projection of (3D) points on a line joining camera centre and a point X

$$\mathbf{P}(\alpha\mathbf{X} + \beta\mathbf{C}) = \alpha\mathbf{PX} = \mathbf{PX}$$

Principal Plane

$$\pi \equiv \mathbf{p}^3$$

Unprojection What points **X** can project to a coordinate **u**?

$$\lambda\mathbf{P}^{+}\mathbf{u} + \mathbf{C}$$

Cameras at Infinity

$$\mathbf{P} = \begin{bmatrix}\mathbf{M} & \mathbf{p}_4\end{bmatrix}$$

rank(**M**) = 2

So there exists **d** $\mathbf{M}\mathbf{d} = \mathbf{0}$

$$\mathbf{P}\begin{bmatrix}\mathbf{d} \\ 0\end{bmatrix} = \mathbf{0} \qquad \mathbf{C} \equiv \begin{bmatrix}\mathbf{d} \\ 0\end{bmatrix}$$

Affine Cameras

$$\mathbf{m}^{3\mathsf{T}} = \mathbf{0}$$

$$\mathbf{P} = \begin{bmatrix}\mathbf{m}^{1\mathsf{T}} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\mathsf{T}} & \mathbf{p}_4^2 \\ \mathbf{0} & \mathbf{p}_4^3\end{bmatrix} = \begin{bmatrix}\mathbf{m}^{1\mathsf{T}} & \mathbf{p}_4^1 \\ \mathbf{m}^{2\mathsf{T}} & \mathbf{p}_4^2 \\ \mathbf{0} & 1\end{bmatrix}$$

Q: Why does the last element have to be non-zero for a valid **P**?

Affine Cameras

$$\mathbf{P} = \begin{bmatrix}\mathbf{M} & \mathbf{p}_4\end{bmatrix}$$

rank(**M**) = 2

So there exists **d** $\mathbf{Md} = \mathbf{0}$

$$\mathbf{P}\begin{bmatrix}\mathbf{d}\\0\end{bmatrix} = \mathbf{0} \qquad \mathbf{C} \equiv \begin{bmatrix}\mathbf{d}\\0\end{bmatrix}$$

Affine Cameras

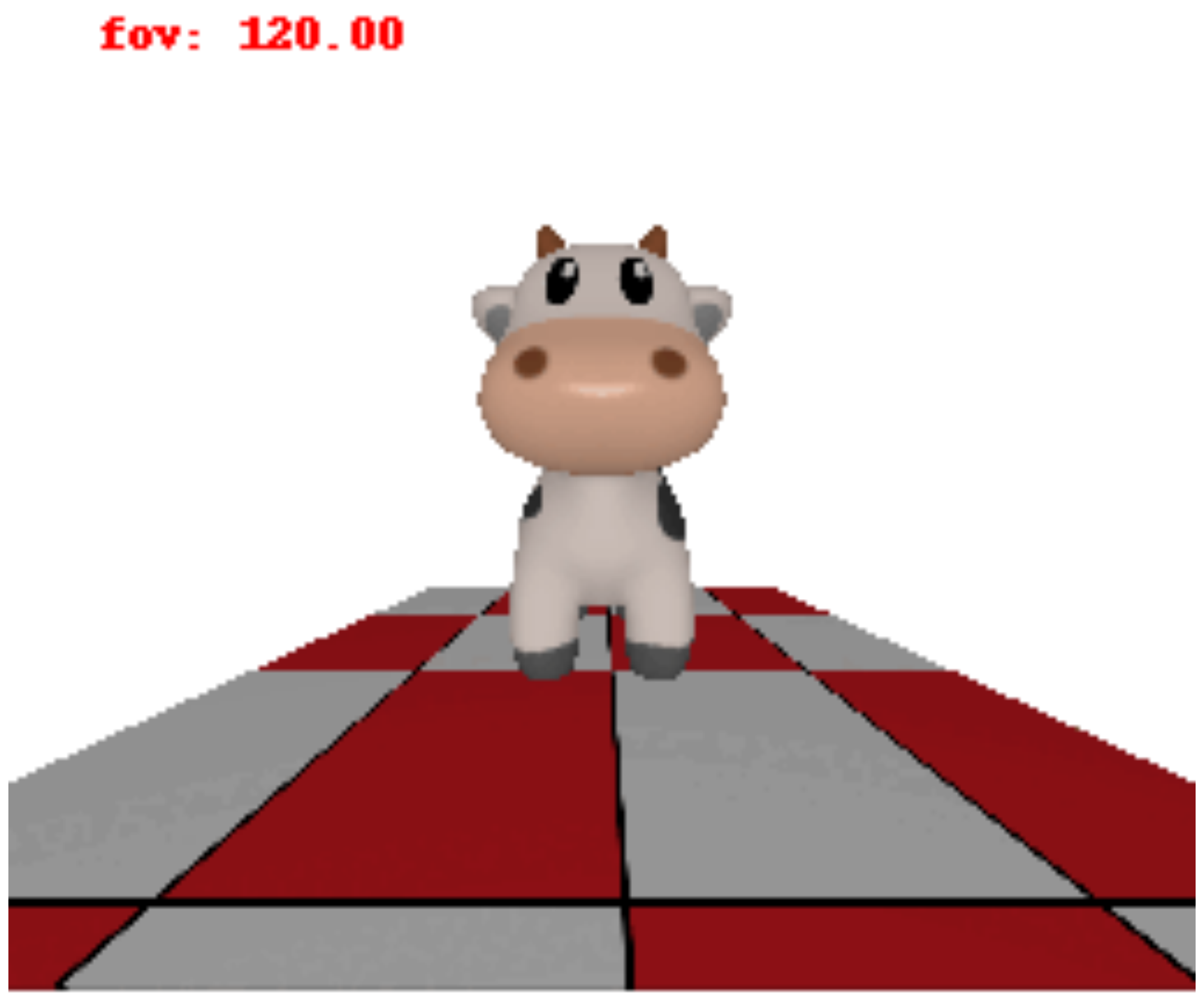
$$\mathbf{m}^{3\top} = \mathbf{0}$$

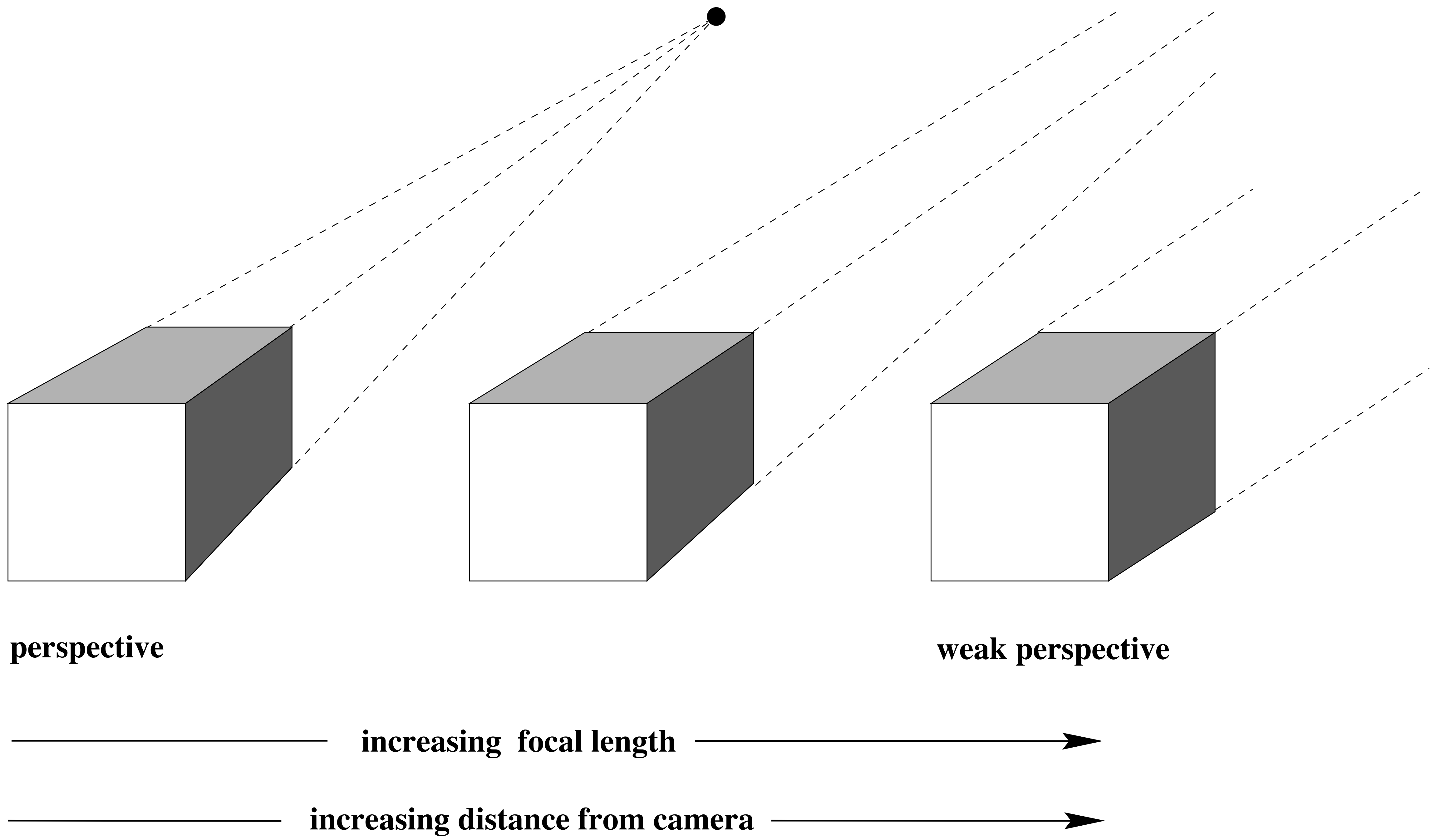
$$\mathbf{P} = \begin{bmatrix}\mathbf{m}^{1\top} & \mathbf{p}_4^1\\ \mathbf{m}^{2\top} & \mathbf{p}_4^2\\ \mathbf{0} & \mathbf{p}_4^3\end{bmatrix} = \begin{bmatrix}\mathbf{m}^{1\top} & \mathbf{p}_4^1\\ \mathbf{m}^{2\top} & \mathbf{p}_4^2\\ \mathbf{0} & 1\end{bmatrix}$$

$$x = \mathbf{m}^{1\top}\tilde{\mathbf{X}} + \mathbf{p}_4^1 \qquad y = \mathbf{m}^{2\top}\tilde{\mathbf{X}} + \mathbf{p}_4^2$$

Image coordinates are an affine transform of the 3D coordinates!

Affine Cameras





Affine camera: infinitely far away,
and infinitely zoomed in

Affine Cameras

$$\mathtt{P}_0 = \mathtt{KR}[\mathtt{I} \mid -\widetilde{\mathbf{C}}] = \mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{3\mathsf{T}} & -\mathbf{r}^{3\mathsf{T}}\widetilde{\mathbf{C}} \end{bmatrix}$$

camera moving along 3rd axis with unit speed

$$\xrightarrow{\qquad\qquad}$$

$$\tilde{\mathbf{C}}_t = \widetilde{\mathbf{C}} - t\mathbf{r}^3$$

$$\mathtt{P}_t = \mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}(\widetilde{\mathbf{C}} - t\mathbf{r}^3) \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}(\widetilde{\mathbf{C}} - t\mathbf{r}^3) \\ \mathbf{r}^{3\mathsf{T}} & -\mathbf{r}^{3\mathsf{T}}(\widetilde{\mathbf{C}} - t\mathbf{r}^3) \end{bmatrix} = \mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{3\mathsf{T}} & d_t \end{bmatrix}$$

+ corresponding zoom

$$\xrightarrow{\qquad\qquad}$$

$$\mathrm{K}_t = \mathtt{K}\begin{bmatrix} d_t/d_0 & & \\ & d_t/d_0 & \\ & & 1 \end{bmatrix}$$

$$\mathtt{P}_t = \mathtt{K}\begin{bmatrix} d_t/d_0 & & \\ & d_t/d_0 & \\ & & 1 \end{bmatrix}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{3\mathsf{T}} & d_t \end{bmatrix} = \frac{d_t}{d_0}\mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{3\mathsf{T}}d_0/d_t & d_0 \end{bmatrix}$$

$$\mathtt{P}_\infty = \lim_{t\to\infty}\mathtt{P}_t = \mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{0}^{\mathsf{T}} & d_0 \end{bmatrix}$$

Affine Cameras

$$\mathtt{K}\begin{bmatrix} \mathbf{r}^{1\top} & -\mathbf{r}^{1\top}\tilde{\mathbf{C}} \\ \mathbf{r}^{2\top} & -\mathbf{r}^{2\top}\tilde{\mathbf{C}} \\ \mathbf{0}^\top & d_0 \end{bmatrix} = \begin{array}{cc} & \text{2 X 1} \\ \multicolumn{2}{c}{\begin{bmatrix} \mathtt{K}_{2\times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^\top & 1 \end{bmatrix}} \end{array} \begin{array}{cc} \text{2 X 3} & \text{2 X 1} \\ \multicolumn{2}{c}{\begin{bmatrix} \hat{\mathtt{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\top & d_0 \end{bmatrix}} \end{array}$$

Q: How many degrees of freedom?

3 + 2 + 3 + 2+ 1 = 11?

$$\mathbf{P} = \begin{bmatrix} K_{2\times 2}\hat{R} & K_{2\times 2}\hat{\mathbf{t}} + d_0\tilde{\mathbf{x}}_0 \\ \mathbf{0} & d_0 \end{bmatrix}$$

Observation 1: Scaling d_0 and K_{2x2} by the same factor does not change P!

Affine Cameras

2 X 1 2 X 3 2 X 1

$$\mathtt{K}\begin{bmatrix} \mathbf{r}^{1\mathsf{T}} & -\mathbf{r}^{1\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{r}^{2\mathsf{T}} & -\mathbf{r}^{2\mathsf{T}}\widetilde{\mathbf{C}} \\ \mathbf{0}^{\mathsf{T}} & d_0 \end{bmatrix} = \begin{bmatrix} \mathtt{K}_{2\times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^{\mathsf{T}} & 1 \end{bmatrix}\begin{bmatrix} \hat{\mathtt{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}$$

Q: How many degrees of freedom?

3 + 2 + 3 + 2 = 10?

$$\mathbf{P} = \begin{bmatrix} K_{2\times 2}\hat{R} & K_{2\times 2}\hat{\mathbf{t}} + \tilde{\mathbf{x}}_0 \\ \mathbf{0} & 1 \end{bmatrix}$$

Observation 2: **t** and $\mathbf{x_0}$ are redundant

Affine Cameras

$$\mathbf{P} = \begin{bmatrix} \mathtt{K}_{2\times 2} & \mathbf{0} \\ \hat{\mathbf{0}}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathtt{R}} & \hat{\mathbf{t}} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix} = \begin{bmatrix} \mathtt{K}_{2\times 2} & \tilde{\mathbf{x}}_0 \\ \hat{\mathbf{0}}^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathtt{R}} & \mathbf{0} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}$$

Equivalent parametrizations

Left one is a more common in code, but remember that **t** is no longer reflective of 3D 'position' of camera!

(rather, it indicates 2D pixel coordinate where world origin projects)

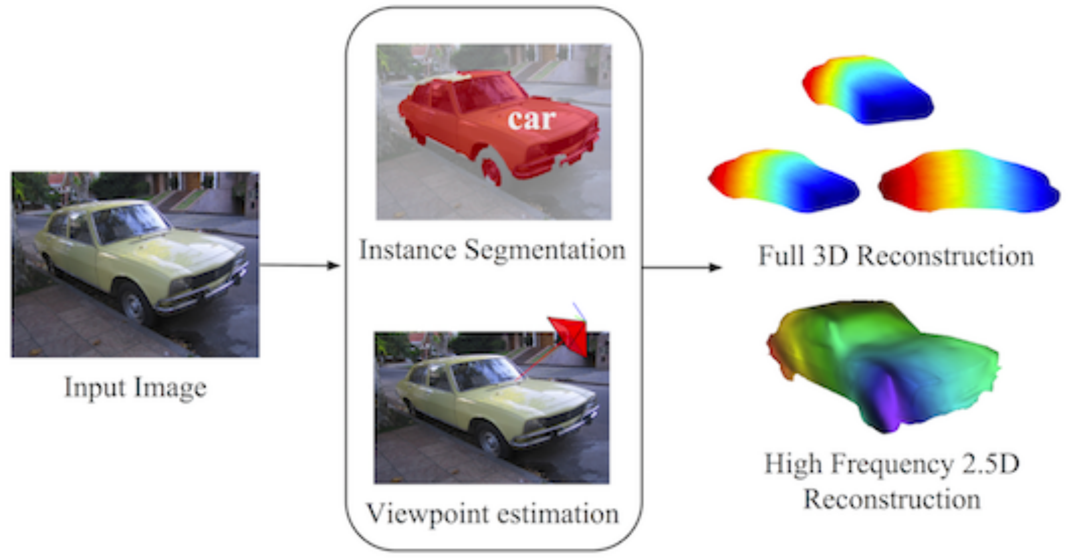
Hierarchy of Affine Cameras

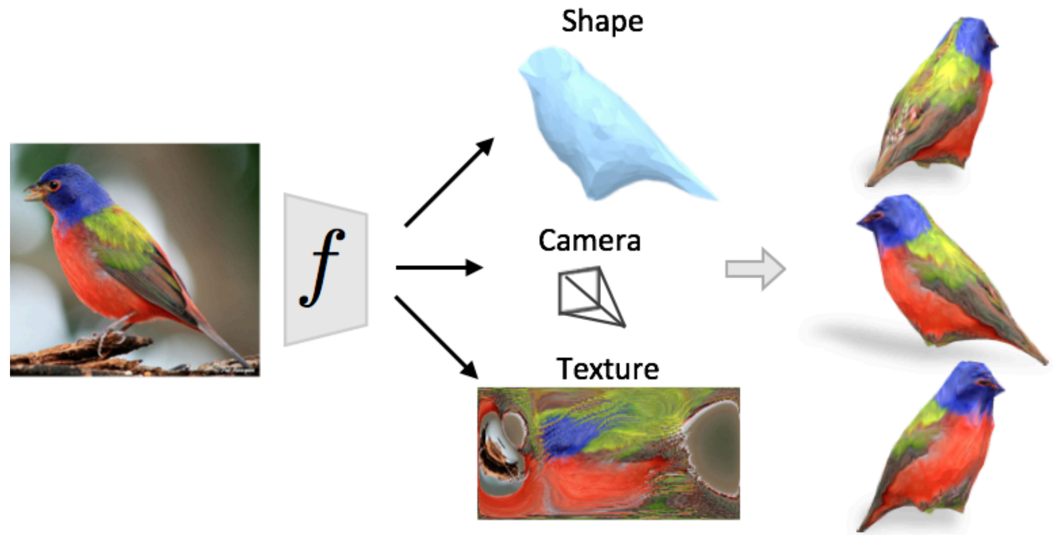
$$\mathbf{P} = \begin{bmatrix} \mathrm{K}_{2\times 2} & \mathbf{0} \\ \hat{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathrm{R}} & \hat{\mathrm{t}} \\ 0^\top & 1 \end{bmatrix} = \begin{bmatrix} \mathrm{K}_{2\times 2} & \tilde{\mathrm{x}}_0 \\ \hat{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathrm{R}} & \mathbf{0} \\ 0^\top & 1 \end{bmatrix}$$

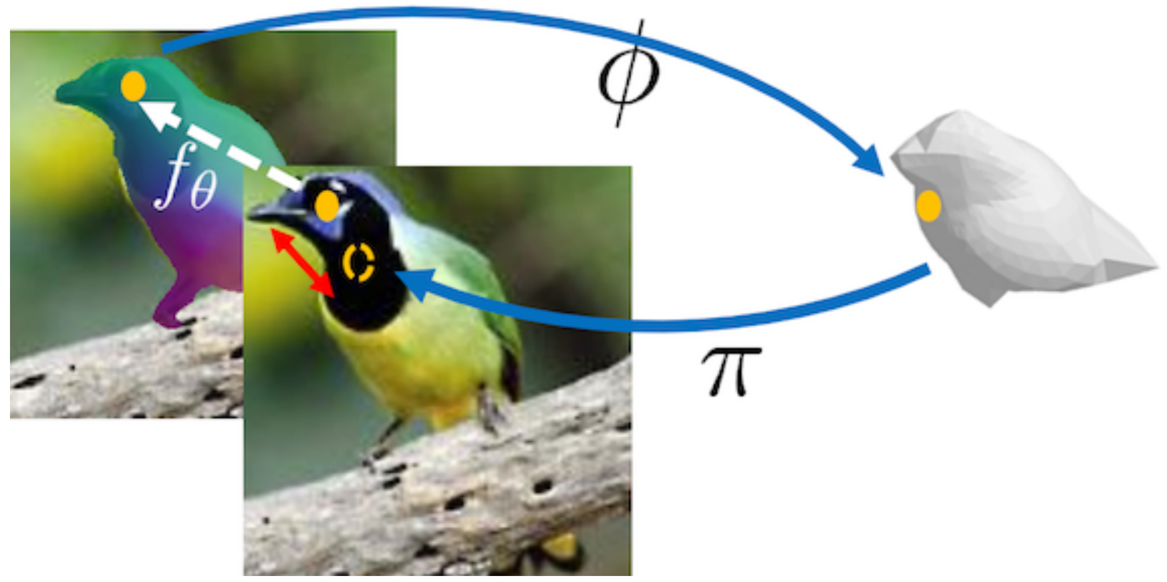
Orthographic: $K_{2\times 2} = I$

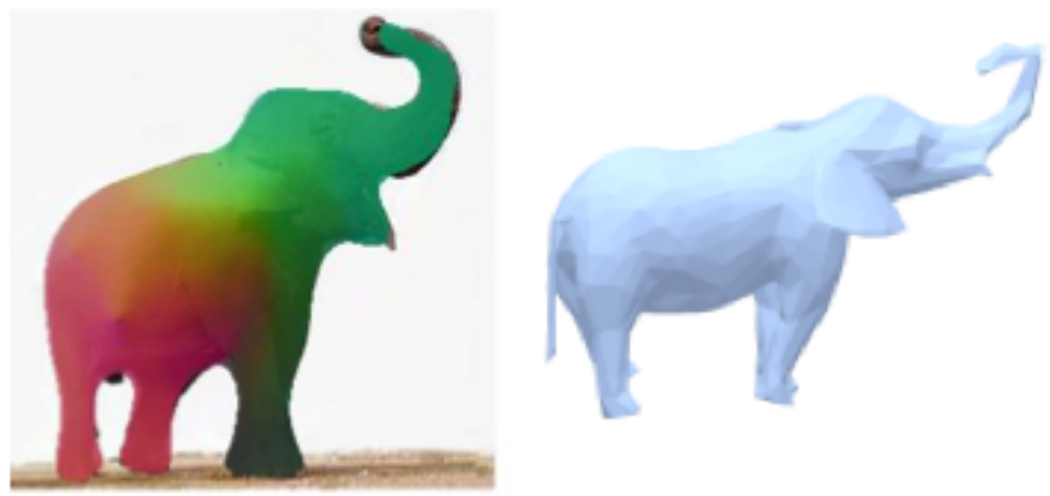
Scaled Orthographic: $K_{2\times 2} = sI$

Weak Perspective: $K_{2\times 2} = \begin{bmatrix} s_1 & 0 \\ 0 & s_2 \end{bmatrix}$



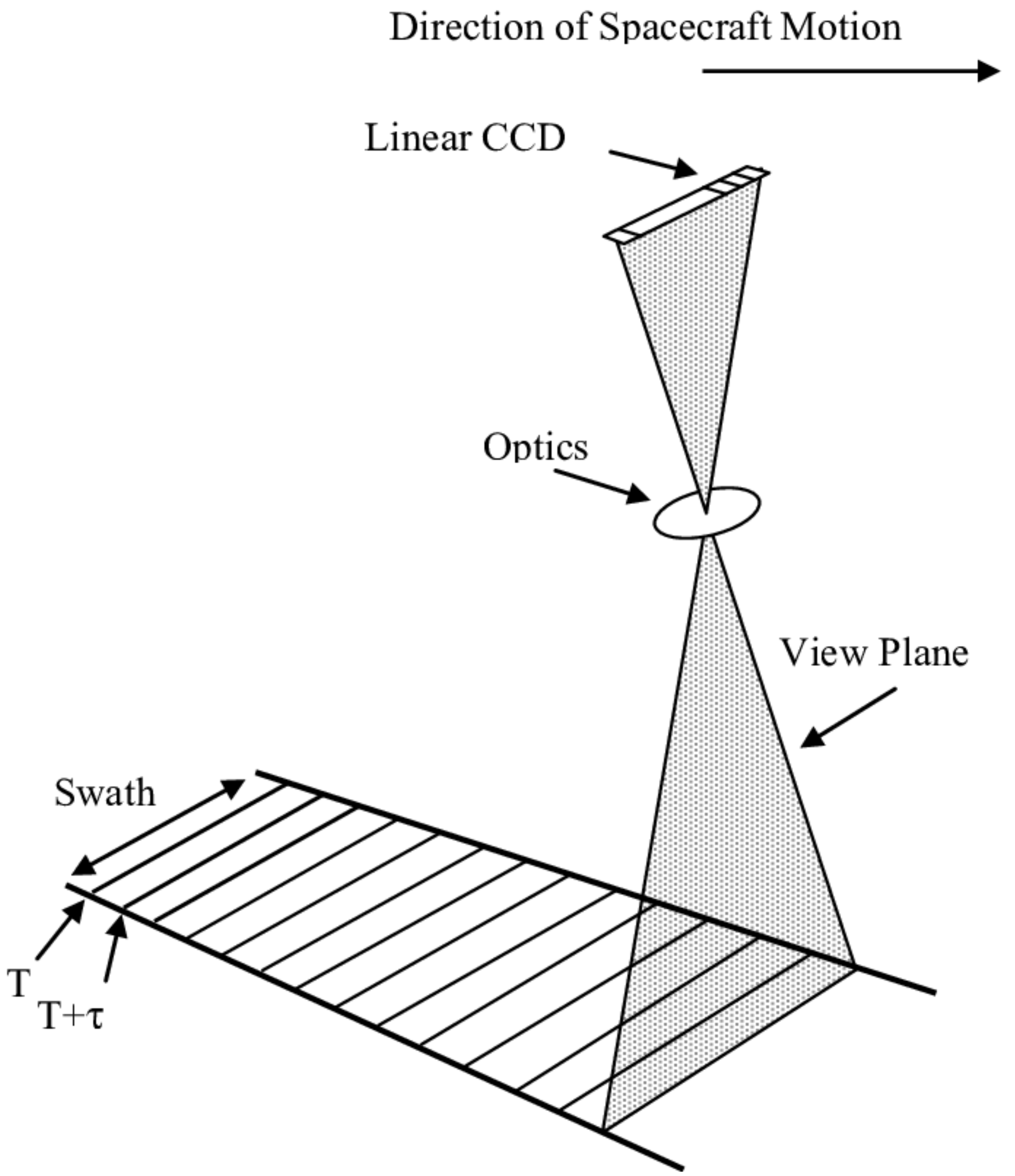






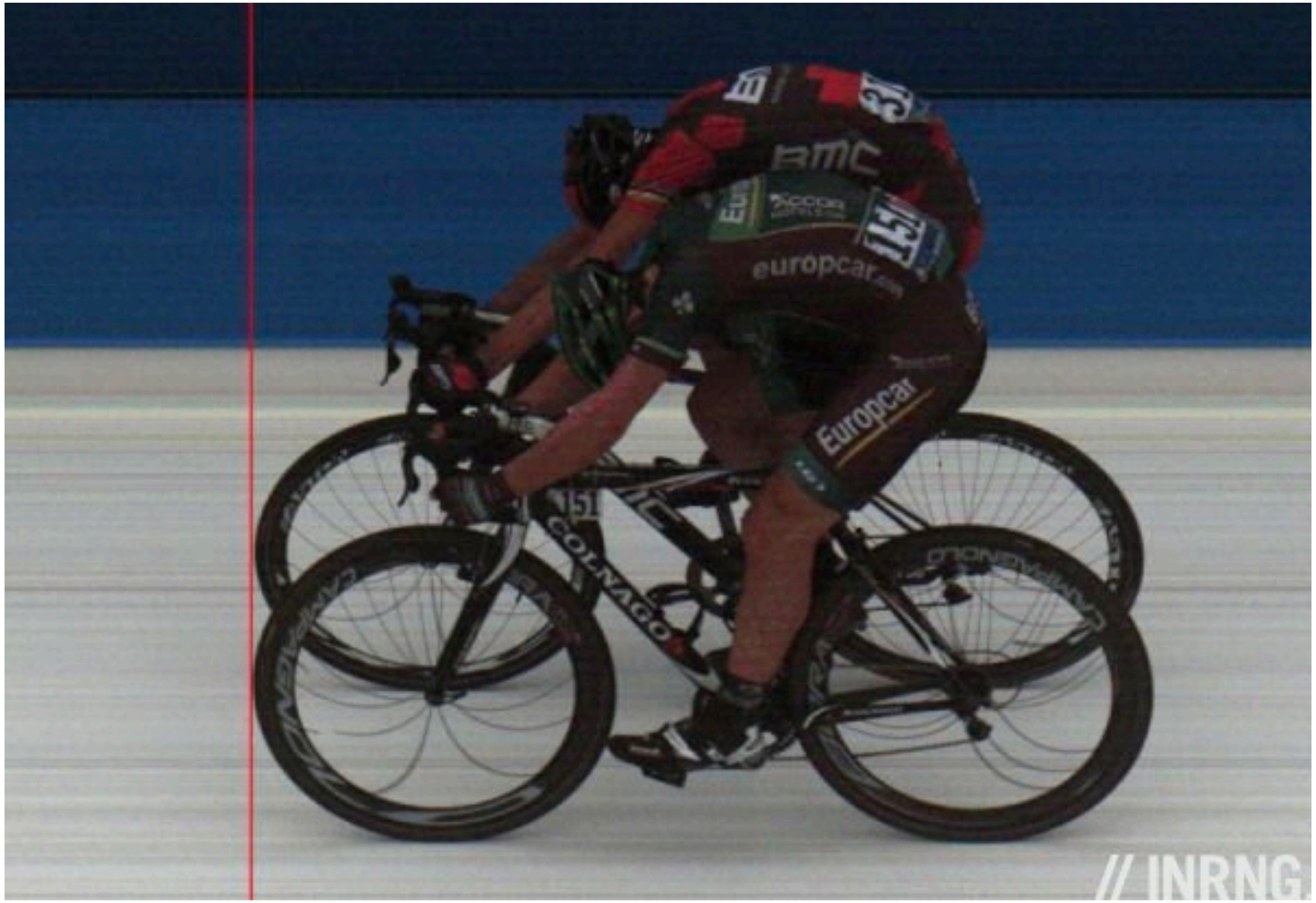
Widely used camera parametrization when learning 3D prediction from images!

Aside: Pushbroom Cameras



$$\mathtt{P}\mathbf{X} = (x, y, w)^{\mathsf{T}}$$

$$(x, y/w)^{\mathsf{T}}$$



Questions?